



Low-Cost EMG Acquisition and Intent Detection for Future Prosthetic Hand Control

by

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April 2026

Abstract

Upper-limb amputation can significantly affect independence, daily function, and quality of life, creating a need for prosthetic technologies that are both responsive and accessible. Although advanced myoelectric prosthetic systems can provide powered hand control, their cost, complexity, calibration requirements, and maintenance demands can limit practical availability. This project investigates a low-cost electromyography (EMG) acquisition and intent-detection system as an early-stage control pathway for future prosthetic hand applications.

The developed system used a SparkFun MyoWare 2.0 muscle sensor and an Arduino Nano ESP32 to acquire forearm muscle activity from surface electrodes. The EMG signal was sampled through the microcontroller, processed using baseline estimation, activation feature extraction, digital smoothing, adaptive thresholding, and logged over USB serial for analysis. A repeated rest/flex protocol was used to evaluate whether deliberate forearm contraction could be distinguished from rest. A 3D-printed DexHand platform was retained as the intended future mechanical output, but full prosthetic hand actuation was not treated as the validated outcome of this work.

The final analysed session recorded 10,401 valid samples at approximately 250 Hz. The cleaned ADC signal ranged from 505 to 583 counts, with a peak filtered activation of 0.02237 V and a mean threshold of 0.01957 V. Limited binary intent detection was achieved, with one of five instructed flex windows detected, zero false triggers outside flex windows, and a measured onset response of 268 ms for the detected trial.

These results show that low-cost EMG acquisition and structured analysis are feasible using accessible hardware. However, the system also demonstrated that reliable intent detection depends strongly on electrode placement, contact quality, cable strain, and signal margin above rest. The work therefore provides a practical foundation for future prosthetic hand control, while identifying the need for improved electrode mounting, calibration, and actuation integration before dependable real-time DexHand control can be achieved.

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Chapter 1 — Introduction

Loss of an upper limb represents one of the most functionally limiting physical impairments, affecting independence, employability, and overall quality of life. Upper-limb function is central to object manipulation, tool use, communication, self-care, and many activities of daily living. As a result, loss of hand or arm function can create long-term physical, psychological, occupational, and social challenges for the affected individual.

Globally, limb loss remains a significant rehabilitation issue, with prevalence influenced by trauma, vascular disease, diabetes, workplace injury, and access to preventative healthcare [1]. In the United States, Ziegler-Graham et al. projected that the number of people living with limb loss would rise from approximately 1.6 million in 2005 to 3.6 million by 2050 [2]. These figures indicate that limb loss is not a static clinical problem, but an increasing long-term healthcare and assistive-technology challenge.

The functional consequences of upper-limb loss are particularly significant. Unlike lower-limb amputation, where mobility aids and prosthetic devices can partially restore locomotion, upper-limb loss directly affects grasping, reaching, dressing, eating, writing, and vocational tasks. Restoring hand function is therefore not only a mechanical problem, but also a human-machine control problem, as the prosthetic device must respond reliably to the user's intended movement.

Rehabilitation following amputation focuses on physical adaptation, motor relearning, prosthetic training, and restoration of independence where possible. Myoelectric prostheses support this goal by using electromyography (EMG) signals from residual or remaining muscles to infer user intent and control powered movement. However, reliable prosthetic control depends on acquiring a stable muscle signal, processing it with low delay, and converting it into a command that can be trusted during everyday use.

This project is positioned within that first stage of the prosthetic control chain: low-cost EMG acquisition and intent detection. Rather than validating a complete clinical prosthetic hand, the work investigates whether accessible hardware can acquire forearm muscle activity, process it in real time, and generate a binary intent signal suitable for future robotic hand control.

Prosthetic Technology Landscape

Prosthetic technology has evolved from passive and body-powered devices toward electrically powered and myoelectric systems. Early upper-limb prostheses commonly relied on cable-driven mechanisms controlled through residual limb, shoulder, or body movement. These systems offered mechanical robustness, relatively low cost, and independence from batteries or embedded electronics. However, they also provided limited dexterity and often required compensatory body movements to perform grasping actions, which could reduce comfort and naturalness during use [3].

Modern myoelectric prostheses use electromyographic (EMG) signals generated during muscle contraction to control powered actuators. By detecting muscle activation patterns in the residual limb or remaining musculature, these systems can support powered hand opening, closing, wrist movement, or selected grip functions without relying entirely on

harness-based mechanical transmission [4]. This makes EMG control an important approach for more intuitive prosthetic interaction, particularly where the aim is to translate user muscle intent into robotic movement.

Despite these advances, dexterous robotic prostheses remain limited by control complexity, cost, weight, calibration requirements, and restricted finger independence. Many commercial and research systems rely on predefined grip modes or switching between control states rather than continuous, natural, independent finger control. This can increase cognitive effort for the user and reduce functional naturalness during daily tasks [4].

Importantly, technological sophistication does not automatically lead to long-term user adoption. Biddiss and Chau reported upper-limb prosthesis rejection rates ranging from approximately 20% to 45%, with users commonly citing discomfort, weight, limited function, poor control, and lack of practical usefulness as reasons for abandonment [5]. These findings show that prosthetic success depends not only on advanced actuation, but also on comfort, reliability, responsiveness, and ease of control.

This project responds to that challenge by focusing on the early sensing and control stage of myoelectric prosthetic design. Before a low-cost robotic hand can be controlled reliably, the system must first acquire a usable EMG signal and convert it into a stable intent output. The work therefore prioritises EMG acquisition and binary intent detection as a foundation for future DexHand actuation.

Cost & Accessibility Barriers

Despite significant technological advancement in upper-limb prosthetics, access to these systems remains constrained by financial, clinical, and infrastructural barriers. The cost of prosthetic provision extends beyond the device itself and can include clinical assessment, socket fabrication, fitting, calibration, rehabilitation training, maintenance, repairs, and replacement components [3].

Body-powered prostheses are generally less complex and less expensive than powered myoelectric systems, but they provide limited dexterity and can require unnatural body movements for control. In contrast, advanced myoelectric prostheses require powered actuators, embedded electronics, batteries, sensors, control software, and specialist clinical fitting. These factors increase both the initial cost and the long-term cost of ownership [3], [6].

Highly dexterous robotic prostheses introduce further cost and accessibility challenges. Multi-articulated hands, individual finger actuation, grip-pattern libraries, and optional sensory feedback systems increase mechanical and electronic complexity. Even when such technologies are technically impressive, their practical use may be limited if they are expensive to obtain, difficult to maintain, or require specialist support that is not widely available [1], [6].

Customisation also contributes to cost. Upper-limb prosthetic devices must be fitted to the user's residual limb geometry, comfort requirements, and functional needs. Socket design and electrode positioning are especially important for myoelectric systems, because poor fit or unstable sensor contact can reduce both comfort and signal reliability. This often requires iterative fitting, training, and adjustment before the device becomes usable in daily life [3].

Maintenance further compounds accessibility challenges. Myoelectric prostheses may require battery replacement, recalibration, component servicing, software updates, and repair of mechanical or electronic parts. These ongoing requirements can limit sustained use, particularly in publicly funded healthcare systems, rural settings, or regions with limited access to specialist prosthetic services [1], [6].

These barriers create a gap between prosthetic technological capability and real-world availability. For this reason, low-cost sensing systems, modular robotic platforms, and open-source prosthetic designs are valuable research directions. This project responds to that need by focusing on an accessible EMG acquisition and intent-detection pathway that could support future low-cost robotic hand control.

Technology Context

To address the functional and accessibility limitations associated with conventional prosthetic systems, research has increasingly focused on control interfaces that can translate user intent into robotic movement. Advances in bio-signal acquisition, wearable sensing, embedded processing, and low-cost robotics have expanded the possibilities for human-machine interaction in assistive and rehabilitation technologies.

Electromyography (EMG) remains one of the most widely used sensing approaches in myoelectric prosthetic control. EMG measures the electrical activity generated during muscle contraction, allowing surface electrodes to provide a practical interface between neuromuscular activation and powered actuation. Myoelectric control systems have been applied to prosthetic arms, robotic gloves, and assistive manipulators, enabling grasp initiation, joint control, and powered movement based on muscle activity [4], [7].

Beyond direct prosthetic control, EMG is also used in gesture-recognition and wearable human-machine interface systems. Multi-channel EMG devices can classify patterns of forearm muscle activation into hand or wrist gestures, supporting robotic control, rehabilitation training, and human-computer interaction applications [8], [9]. However, EMG signals are sensitive to electrode placement, skin impedance, motion artefacts, cable movement, and muscle fatigue. These factors can reduce repeatability and make reliable control difficult without careful filtering, calibration, and sensor mounting.

Inertial measurement units (IMUs) provide a complementary form of sensing by measuring movement and orientation rather than muscle activation. IMUs use accelerometers, gyroscopes, and sometimes magnetometers to estimate limb motion, segment orientation, and angular velocity. In rehabilitation and teleoperation research, IMU-based systems have been used to estimate joint angles, track limb movement, and support robotic arm mapping without relying on external camera systems [10], [11].

Because EMG and IMU sensing capture different aspects of user behaviour, hybrid systems have received increasing research attention. EMG can provide information about muscle intent, while IMUs provide kinematic context such as arm orientation or movement. Wearable systems combining EMG and IMU signals have shown improved gesture-recognition performance and richer control information compared with single-modality approaches [8], [12].

In the context of this project, EMG was selected as the validated sensing pathway because detecting muscle activation is the first requirement for future myoelectric hand control. IMU sensing remains relevant as a future extension for arm orientation tracking or hybrid control, but it was not implemented as part of the final validated experimental system. The project therefore focuses on low-cost EMG acquisition, filtering, logging, and binary intent detection as the foundation for later prosthetic hand integration.

Research Gap

Despite significant advancements in prosthetic actuation, bio-signal sensing, and robotic rehabilitation systems, reliable and accessible myoelectric control remains a major challenge. Existing upper-limb prosthetic interfaces can be limited by signal variability, electrode placement sensitivity, calibration burden, cost, and difficulty maintaining stable control during everyday use [3], [5].

Myoelectric prosthetic systems are capable of translating muscle activation into powered movement, but practical control is often constrained by noisy or inconsistent EMG signals. Surface EMG is affected by skin contact, electrode position, muscle fatigue, motion artefacts, and changes in posture. As a result, even before complex gesture classification or robotic actuation is considered, the system must first acquire a usable and repeatable muscle signal [4], [7].

Many research systems focus on multi-channel gesture recognition, hybrid EMG-IMU sensing, or advanced robotic actuation. These approaches are valuable, but they can also introduce additional hardware, calibration, synchronisation, and processing complexity [8], [12]. For low-cost prosthetic development, there remains a need to validate the simpler first stage of the control chain: whether accessible hardware can reliably acquire forearm EMG activity, filter it, log it, and convert it into a basic intent signal.

A further gap exists between laboratory demonstrations and practical low-cost implementation. Published systems often report classification accuracy or teleoperation performance, but less attention is sometimes given to the practical setup issues that affect repeatability, such as electrode adhesion, cable strain, baseline drift, threshold selection, and false triggering during rest. These issues are especially important for rehabilitation-oriented or student-built systems, where affordability and ease of setup are central design constraints.

This project addresses that gap by developing and evaluating a low-cost EMG acquisition and binary intent-detection workflow using a MyoWare 2.0 sensor and Arduino Nano ESP32. Rather than claiming full prosthetic hand control, the project focuses on the foundational sensing stage required before reliable DexHand actuation can be achieved. The work evaluates signal acquisition, filtering, thresholding, serial logging, missed detections, false triggers, and practical setup limitations under a repeated rest/flex protocol.

Aim & Objectives

Aim

The aim of this project is to develop and evaluate a low-cost EMG acquisition and intent-detection system for future prosthetic hand control. The work focuses on acquiring forearm muscle activity using accessible hardware, processing the signal in real time, and determining whether deliberate contraction can be converted into a binary intent output suitable for later robotic hand integration.

The project does not claim to validate a complete clinical prosthetic hand, full DexHand actuation, or a hybrid EMG-IMU robotic arm system. Instead, it evaluates the foundational sensing and processing stage required before dependable prosthetic actuation can be achieved.

Objectives

To achieve this aim, the following objectives were defined:

1. EMG Signal Acquisition

- Design and implement a single-channel surface EMG sensing system using a MyoWare 2.0 muscle sensor and disposable surface electrodes.
- Connect the EMG sensing hardware to an Arduino Nano ESP32 for embedded analog acquisition.
- Investigate practical electrode placement and skin-contact conditions for forearm muscle activation detection.

2. Embedded Signal Processing

- Sample the EMG envelope signal using the Nano ESP32 analog input.
- Implement baseline estimation, activation feature extraction, smoothing, adaptive thresholding, and hysteresis-based binary intent detection.
- Generate a real-time intent output representing rest or detected contraction.

3. Serial Logging and Experimental Protocol

- Develop a host-side logging workflow to capture EMG data over USB serial.
- Implement a guided repeated rest/flex protocol to create repeatable contraction windows for analysis.
- Record sample data, marker events, and processed signal values for offline evaluation.

4. Data Analysis and Evaluation

- Analyse captured EMG sessions using plots, summary metrics, and threshold-crossing results.
- Evaluate signal separation, missed detections, false triggers, and onset response during repeated contraction trials.
- Identify the practical factors limiting repeatability, including electrode placement, cable movement, and signal margin above rest.

5. Future Prosthetic Integration Context

- Use the 3D-printed DexHand platform as the intended future mechanical target for EMG-driven actuation.
- Identify the hardware and control requirements needed for future servo-driven DexHand movement.
- Clearly separate validated EMG acquisition results from future actuation, IMU fusion, and multi-channel gesture-control work.

Collectively, these objectives support the development of a practical low-cost EMG control foundation for future prosthetic hand research, while keeping the validated project claims aligned with the experimental evidence collected.

Scope & Limitations

This project focuses on the design, development, and technical evaluation of a low-cost EMG acquisition and intent-detection system for future prosthetic hand control. The work is positioned within assistive robotics and myoelectric control, with emphasis placed on signal acquisition, embedded processing, serial logging, and threshold-based intent detection rather than clinical deployment or fully validated robotic hand actuation.

The validated scope of the project includes a single-channel surface EMG sensing interface using a MyoWare 2.0 muscle sensor, disposable surface electrodes, and an Arduino Nano ESP32. The system acquires forearm muscle activation, estimates a baseline, extracts an activation feature, applies smoothing and thresholding, and outputs a binary rest/contraction intent state. A guided repeated rest/flex protocol is used to evaluate signal response, missed detections, false triggers, and onset behaviour.

The DexHand platform is included as the intended future mechanical output and as a physical demonstration of the prosthetic-hand application context. However, full DexHand actuation, individual finger control, proportional grip control, and clinical prosthetic validation are outside the validated scope of this project. IMU sensing and robotic arm teleoperation were considered during the wider concept stage, but they are treated as future extensions rather than implemented experimental outcomes.

The project should therefore be understood as an EMG-first feasibility study. Its main contribution is the development and evaluation of the low-cost sensing and intent-detection stage required before more complex prosthetic hand control can be attempted.

Limitations

While the system is developed with future rehabilitation and prosthetic applications in mind, several limitations define the boundaries of the present work.

Firstly, experimental testing was conducted as a single-participant feasibility study using the project author. No clinical trials involving amputees, prosthesis users, or neurologically impaired participants were undertaken. As such, conclusions regarding prosthetic usability, therapeutic effectiveness, or clinical reliability cannot be made from this work.

Secondly, the validated system used a single EMG channel and a binary intent output. It did not validate multi-channel gesture recognition, proportional grip control, independent finger control, or continuous prosthetic movement. The binary output should therefore be interpreted as an early control signal rather than a complete prosthetic control strategy.

Thirdly, the DexHand platform was used as the intended future mechanical target and project demonstration context rather than as a medically certified prosthetic device. Full DexHand actuation was not validated as the main experimental outcome. Mechanical comfort, socket design, long-term wearability, force feedback, tactile sensing, and medical safety certification were outside the scope of this investigation.

Additionally, surface EMG performance was strongly affected by practical setup conditions, including electrode placement, skin contact, cable strain, and signal margin above rest. Testing was conducted under structured rest/flex conditions, so long-term electrode wear, sweat-induced signal degradation, movement during daily activities, and environmental interference were not exhaustively evaluated.

Finally, cost analysis within this work was limited to component-level prototyping considerations. Full clinical deployment costs, including socket fabrication, rehabilitation training, regulatory approval, maintenance, and long-term support, were not examined.

Despite these limitations, the project provides a technically grounded foundation for low-cost EMG acquisition and intent detection. It identifies the practical sensing, filtering, and calibration challenges that must be addressed before reliable DexHand actuation or future prosthetic hand control can be achieved.

Thesis Structure

This thesis is structured to present the project from background theory through methodology, implementation, experimental evaluation, and future development.

Chapter 1 — Introduction

Chapter 1 introduces the context of upper-limb prosthetics, myoelectric control, accessibility barriers, and the motivation for low-cost EMG acquisition. It defines the research gap, aim, objectives, scope, and limitations of the project.

Chapter 2 — Literature Review

Chapter 2 reviews human-machine interface technologies relevant to assistive robotics, including EMG-based control, IMU-based motion tracking, hybrid EMG-IMU systems, vision-based interfaces, and brain-computer interfaces. The chapter compares these approaches in terms of accuracy, latency, robustness, ease of use, and rehabilitation value, while identifying the need for reliable low-cost EMG acquisition as a foundation for future prosthetic control.

Chapter 3 — Methodology

Chapter 3 describes the methodology used to develop and evaluate the EMG acquisition and intent-detection system. It explains the hardware selection, electrode placement approach, signal processing pipeline, repeated rest/flex protocol, and performance metrics used to assess the system.

Chapter 4 — System Design and Implementation

Chapter 4 presents the implemented hardware, firmware, logging workflow, and analysis pipeline. It also describes the DexHand platform as the intended future mechanical target, while distinguishing between the validated EMG acquisition system and future actuation work.

Chapter 5 — Experimental Setup and Results

Chapter 5 presents the experimental setup, guided rest/flex testing procedure, captured EMG sessions, signal plots, and measured results. The chapter evaluates raw signal response, filtered threshold behaviour, missed detections, false triggers, and the limited binary intent detection achieved in the final analysed session.

Chapter 6 — Discussion

Chapter 6 interprets the results in relation to low-cost EMG acquisition and future prosthetic hand control. It discusses electrode placement sensitivity, contact quality, thresholding limitations, DexHand mechanical lessons, and the practical requirements for improving repeatability.

Chapter 7 — Conclusion and Future Work

Chapter 7 summarises the project outcomes and reflects on the effectiveness of the developed EMG acquisition workflow. It identifies future work including improved electrode mounting, multi-channel EMG, improved calibration, positional servo actuation, DexHand integration, and optional IMU fusion.

Chapter 2 — Literature Review

Chapter Outline

1. *Introduction*

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- *Assistive robotics and future prosthetic hand control*
- *EMG as the primary sensing focus*
- *IMU, vision, BCI, and hybrid systems as wider context*

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- *Latency*
- *Robustness and repeatability*
- *Ease of use*
- *Rehabilitation and prosthetic relevance*

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- *Robot teleoperation and environmental awareness*
- *Strengths and limitations in rehabilitation contexts*

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- *Muscle-intent detection and prosthetic control*
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- *Orientation tracking and sensor fusion*
- *Drift, calibration, and mounting limitations*

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- *Assistive robot control applications*
- *Practical deployment barriers*

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- *Physiological versus kinematic information*
- *Simplicity versus control richness*
- *Low-cost implementation versus reliability*

7. Identified Research Gap

- *Need for reliable low-cost EMG acquisition*
- *Limited focus on repeatability, thresholding, and false triggers*
- *Practical gap between laboratory systems and accessible prosthetic prototypes*
- *Proposed project contribution*

8. Summary

- *Trends in interface development*
- *Shared weaknesses across sensing modalities*
- *Motivation for an EMG-first, low-cost intent-detection system*

1. Introduction

This literature review examines human-machine interface technologies relevant to assistive robotics and future prosthetic hand control, with particular emphasis on electromyography (EMG) as a practical sensing method for detecting muscle intent. The review also considers complementary sensing and control approaches, including inertial measurement units (IMUs), vision-based systems, brain-computer interfaces, and hybrid EMG-IMU architectures. These technologies are reviewed to understand how user intent can be detected, processed, and translated into robotic movement.

EMG-based control is especially relevant to this project because it provides a direct method of measuring muscle activation through surface electrodes. Previous work has shown that EMG signals can be used to control prosthetic or assistive robotic systems, support power-grasp detection, and classify hand or wrist gestures [4], [7], [9]. However, EMG control is also affected by practical limitations such as electrode placement, signal noise, fatigue, calibration requirements, and variability between users.

IMU-based systems and hybrid EMG-IMU approaches are also reviewed because they provide useful context for future development. IMUs can track limb orientation and motion, while EMG provides information about muscle intent. Combining these signals can improve recognition and control richness in some applications, but it also increases hardware complexity, synchronisation requirements, and calibration burden [8], [10], [11], [12].

The purpose of this review is therefore not to justify a fully validated hybrid control system in the present project. Instead, it identifies the wider technology landscape and shows why a reliable low-cost EMG acquisition stage is an important foundation for later prosthetic hand control. Existing research is compared in terms of accuracy, latency, robustness, ease of use, and rehabilitation value, with attention given to the practical limitations that affect real-world implementation.

This analysis supports the research direction of the present project: developing and evaluating a low-cost EMG acquisition and binary intent-detection workflow before attempting more complex DexHand actuation, multi-channel gesture recognition, or IMU-based arm tracking.

2. Overview of Human–Machine Interfaces in Assistive Robotics

Human-machine interfaces in assistive robotics use sensing technologies to interpret user intention and convert it into robotic commands. Different modalities capture different forms of information. Vision systems observe external motion, EMG measures muscle activation, IMUs track movement and orientation, and brain-computer interfaces attempt to infer intent from neural activity. This section reviews these interface types to compare their strengths, limitations, and relevance to rehabilitation-oriented robotic control.

2.1 Vision-Based Interfaces

Vision-based interfaces use cameras to interpret user movement, gestures, posture, or object interaction and convert this information into robot commands. Webcam-based hand gesture recognition has been used to control a transporter robot in real time, demonstrating that low-cost visual sensing can support intuitive device-free interaction [13]. Other work has used RGB-D camera tracking to follow wrist trajectories for robot teleoperation, allowing a robot

arm to reproduce user movement even when visual data are noisy or sampled at a relatively low rate [14]. Vision systems are also used in industrial robotic applications, where eye-in-hand cameras can support object detection and pick-and-place control [15].

A major advantage of vision-based interfaces is that the user does not need to wear sensors, electrodes, or markers. This makes them attractive for applications where natural interaction and fast setup are important. However, vision systems are sensitive to lighting conditions, camera viewpoint, occlusion, background clutter, and visual noise. These issues can reduce reliability, especially when the user's hand or arm is partially hidden or moving quickly.

For upper-limb rehabilitation and prosthetic control, vision-based interfaces are useful for gross movement observation, teleoperation, and environmental awareness. However, they are less suitable as the primary control method for detecting muscle intent or fine grasp initiation. Unlike EMG, vision cannot directly measure neuromuscular activation before movement occurs. This limitation makes vision valuable as a complementary sensing method, but less appropriate for the EMG-first control pathway investigated in this project.

2.2 EMG-Based Interfaces

Electromyography (EMG) interfaces measure the electrical activity generated by muscle fibres during contraction. Surface EMG electrodes placed on the skin can capture activation patterns from underlying muscles, providing a practical method of detecting user intent before or during visible movement. This makes EMG especially relevant to myoelectric prostheses, robotic gloves, and assistive devices that aim to respond to voluntary muscle activity [4], [7].

In robotic control, EMG signals are commonly processed using amplitude, envelope, or extracted feature values and then mapped to actuator commands. Some systems use EMG to control joint movement or prosthetic arm functions, while others classify hand and wrist gestures using multi-channel EMG or wearable armband systems [4], [8], [9]. Single-sensor EMG approaches have also been investigated for power-grasp detection, showing that simple EMG configurations can support useful assistive control when the sensing location is carefully selected [7].

The main advantage of EMG is that it provides direct access to muscle intent. This is valuable for prosthetic control because the system can respond to the user's own neuromuscular activation rather than relying only on visible motion. EMG is therefore well suited to the present project, which focuses on acquiring forearm muscle activity and converting it into a binary intent signal for future robotic hand control.

However, EMG-based interfaces also face major practical challenges. Signal quality can be affected by electrode placement, skin impedance, motion artefacts, cable movement, sweat, muscle fatigue, and cross-talk from nearby muscles. These factors can reduce repeatability and may require filtering, calibration, secure electrode mounting, and careful threshold selection [4], [5], [7]. These limitations directly relate to the present project, where electrode contact and signal margin were found to be key constraints.

2.3 IMU-Based Interfaces

Inertial measurement unit (IMU) interfaces use accelerometers, gyroscopes, and sometimes magnetometers to measure movement and orientation. These sensors provide time-series data describing acceleration, angular velocity, and segment orientation, making them useful for tracking limb movement in rehabilitation, activity recognition, and robot teleoperation [17], [18].

IMUs have been used in motion-capture systems to estimate joint angles and compare movement against camera-based systems [10]. They have also been applied in robot teleoperation, where sensor-fusion methods such as the Madgwick filter are used to estimate orientation and transmit movement information to a robotic arm controller [11]. Compared with camera-based systems, IMUs are lightweight, wearable, relatively low cost, and not affected by visual occlusion or lighting conditions.

However, IMU systems also have limitations. Orientation estimates can drift over time, magnetometers can be affected by nearby metal or electromagnetic interference, and poor sensor placement can reduce accuracy. Sensor movement on the body can also affect repeatability, especially during longer sessions [17], [18]. For this project, IMUs are relevant as a future extension for arm orientation tracking, but they were not part of the final validated experimental system.

2.4 Brain-Computer Interfaces

Brain-computer interfaces (BCIs) measure neural activity, typically through scalp electroencephalography (EEG), and attempt to translate that activity into control commands. BCI systems have been investigated for robot control, rehabilitation support, and assistive communication, particularly for users with severe motor impairments who may not be able to use conventional muscle-based interfaces [18], [19], [20].

The main advantage of BCIs is that they can provide a control pathway that does not depend on residual limb movement or muscle activation. This makes them potentially valuable for users with very limited motor function. General-purpose platforms such as BCI2000 demonstrate how neural signals can be acquired, processed, and converted into device commands in real time [21].

Despite this potential, BCIs remain difficult to deploy reliably outside controlled environments. EEG signals are low amplitude, sensitive to noise and artefacts, and often require significant setup, training, and signal processing. Wireless and implantable systems also introduce hardware, power, safety, and reliability challenges [21]. For these reasons, BCIs are important in the wider assistive-technology landscape, but they are beyond the scope of the present low-cost EMG acquisition project.

3. EMG Control of Dexterous Robotic Hands

EMG-based control of robotic hands aims to translate muscle activation into useful prosthetic or assistive movement. In advanced systems, this may involve multi-gesture classification, proportional control, or independent finger actuation. However, all of these approaches depend on the same foundational requirement: acquiring a stable and repeatable EMG signal from the user. This section reviews EMG acquisition, feature extraction, classification, control strategies, and performance metrics relevant to dexterous robotic hand control.

3.1 EMG Signal Acquisition

EMG-based hand-control systems typically use surface electrodes placed on the skin to detect muscle activation. For upper-limb applications, electrodes are commonly positioned over forearm flexor and extensor muscle groups because these muscles are associated with grasping, wrist motion, and finger movement [4], [7]. The exact placement is important because small changes in electrode position can affect signal amplitude, noise, and cross-talk from nearby muscles.

Some robotic and prosthetic systems use multiple EMG channels to capture different muscle activation patterns. For example, low-cost robotic arm controllers may position electrodes over antagonist muscle groups such as the flexor carpi radialis and extensor digitorum to distinguish between different movement intentions [17]. Other systems investigate simpler single-sensor configurations, where targeted placement can detect power-grasp intent using only one EMG sensor [7].

Hardware used in published work ranges from commercial EMG modules and wearable armbands to custom multi-channel acquisition systems. Multi-channel systems provide richer information for gesture recognition, but they also increase setup complexity, calibration requirements, and cost. In contrast, single-channel systems are less capable of detailed gesture separation but can be useful for investigating basic rest-versus-contraction detection. This makes single-channel EMG appropriate for early-stage low-cost prosthetic control research, including the present project.

3.2 Feature Extraction and Classification

Raw EMG signals are usually processed before they are used for control. Common feature extraction methods include root mean square (RMS), mean absolute value (MAV), zero-crossing count, waveform length, and slope sign changes. These features help describe signal amplitude, frequency-related behaviour, and changes in muscle activation over time. Some systems also use slope-based or envelope-based features to estimate contraction strength or movement intent [4].

For gesture recognition, EMG features are often passed into classifiers such as support vector machines, artificial neural networks, k-nearest neighbour methods, linear discriminant analysis, or other machine-learning approaches. Multi-channel sEMG systems can report high gesture-recognition accuracy under controlled laboratory conditions, particularly when the number of gestures, electrode placement, and participant setup are well controlled [9].

However, classification accuracy reported in the literature does not always transfer directly to real-world prosthetic or rehabilitation use. EMG signals can change due to electrode shift,

skin impedance, sweat, fatigue, posture, and user variation. As a result, even accurate classifiers may require recalibration or may perform less reliably outside controlled test conditions [7], [9]. For this reason, the present project focuses first on acquiring and thresholding a usable signal before attempting multi-gesture classification.

3.3 Control Strategies

EMG-based robotic hand control commonly uses either classification-based control or proportional control.

Classification-based control maps EMG features to discrete commands such as open, close, pinch, or wrist rotate. The system identifies the most likely intended gesture and triggers the corresponding robotic action. This approach is common in gesture-recognition systems and can work well for small predefined command sets [9]. However, it can become less intuitive when many modes must be selected or switched during daily use.

Proportional control maps EMG signal magnitude to actuator speed, grip force, or finger displacement. In principle, this allows the user to produce stronger or weaker robotic movement by varying muscle contraction intensity. This approach can feel more natural than discrete switching, but it is more sensitive to noise, fatigue, baseline drift, and calibration quality [4].

Independent finger control remains particularly challenging because forearm muscles overlap in function and individual finger movements can produce similar or coupled EMG patterns. Multi-channel systems and additional sensors can improve separation, but they also increase complexity [4]. For low-cost systems, a simpler binary grasp-intent signal may be a more realistic first stage before attempting proportional or individual-finger control.

3.4 Performance Metrics

EMG control systems are commonly evaluated using metrics such as classification accuracy, latency, robustness, and repeatability.

Accuracy describes how often the system correctly identifies the intended gesture or control state. Multi-channel gesture-recognition systems can report high accuracy under controlled conditions, while single-channel approaches are usually more limited but may still be suitable for simple binary tasks such as detecting grasp intent [7], [9].

Latency describes the delay between user muscle activation and system response. It is affected by sampling rate, filtering, feature-window length, classifier complexity, communication delay, and actuator response. In prosthetic or assistive control, excessive latency can make the device feel unnatural or harder to trust. For this reason, low-latency embedded processing is important, especially when the control output is intended to drive real-time hand movement [4], [9].

Robustness and repeatability are equally important. EMG systems must cope with electrode movement, skin contact changes, fatigue, and environmental noise. Muscle fatigue can alter EMG amplitude and frequency characteristics over time, reducing classification stability and proportional-control reliability [23]. For rehabilitation or repeated-use applications, a system

must therefore remain stable across multiple contractions, not just perform well during a single short trial.

These metrics directly informed the present project. Rather than evaluating complex gesture accuracy, the project evaluates whether a low-cost single-channel EMG system can produce a measurable rest-versus-flex response, avoid false triggers, and generate a limited binary intent output under a repeated rest/flex protocol.

4. IMU-Based Control of Robotic Arms

IMU-based control systems use inertial sensing to estimate human limb movement and map that movement onto robotic devices. In upper-limb rehabilitation and teleoperation, IMUs can provide information about arm orientation, joint movement, and motion trajectories. Although IMU control was not implemented in the final validated project, it remains relevant as a future extension because it could provide arm-position context alongside EMG-based grasp intent.

4.1 Motion Sensing Fundamentals

Inertial measurement units typically combine accelerometers, gyroscopes, and sometimes magnetometers to measure acceleration, angular velocity, and orientation. When mounted on body segments such as the upper arm, forearm, or wrist, IMUs can be used to estimate movement patterns and reconstruct aspects of limb posture [17], [18].

In teleoperation studies, MARG sensors, which combine magnetic, angular-rate, and gravity sensing, have been used to capture arm orientation and support robotic arm control. Sensor-fusion methods can convert raw inertial data into orientation estimates that are more stable than using individual sensor outputs alone [11]. Motion-capture research has also shown that wearable IMU systems can estimate joint angles and provide an alternative to camera-based tracking systems [10].

The types of movement tracked by IMUs include segment orientation, flexion-extension, rotation, angular velocity, and acceleration. These measurements are useful for robotic arm mapping because they provide kinematic information that EMG alone cannot provide. While EMG detects muscle activation, IMUs describe how the limb is moving in space.

4.2 Mapping Human Motion

Human motion can be mapped to robotic movement by converting IMU-derived orientation data into joint-angle estimates or motion commands. Sensor-fusion algorithms such as the Madgwick filter combine gyroscope, accelerometer, and magnetometer data to estimate orientation in real time [11]. These orientation estimates can then be used within a simplified kinematic model of the human arm.

In teleoperation systems, human joint angles may be mapped directly to robot joint commands, scaled to match the robot's range of motion, or processed through inverse kinematics to reproduce an end-effector trajectory. For rehabilitation applications, this approach can support motion tracking, exercise monitoring, or active-assist robotic movement.

However, mapping human motion to a robotic arm is not straightforward. Human joints and robot joints may not align perfectly, sensor placement can shift during use, and calibration is required to define the starting orientation. These issues can affect repeatability and user comfort, particularly during longer sessions.

4.3 Reported Performance and Limitations

Performance in IMU-based control is commonly evaluated using tracking error, delay, drift, and usability. Motion-capture comparisons have reported joint-angle errors within a few degrees for some movements when sensors are correctly placed and calibrated [10]. IMU-based teleoperation studies using sensor-fusion filtering can also support real-time robotic control, although performance depends on filtering, communication, and calibration quality [11].

IMU systems offer several practical advantages. They are lightweight, wearable, relatively low cost, and not dependent on lighting conditions or external cameras. This makes them attractive for portable rehabilitation systems and robotic teleoperation [17], [18].

Despite these advantages, IMU systems face important limitations. Gyroscope drift can accumulate over time, magnetometers can be disturbed by nearby metal or electromagnetic interference, and poor sensor attachment can reduce accuracy. Sensor misalignment, vibration, and body movement outside the intended plane can also introduce errors. For this reason, IMUs require careful mounting and calibration before they can provide reliable control information.

For the present project, IMU-based control is best understood as a future development path rather than a validated result. A future system could combine EMG-based grasp intent with IMU-based arm orientation tracking, but the final validated thesis work focuses on the EMG acquisition and binary intent-detection stage.

5. Hybrid EMG-IMU Systems and Control Architecture Considerations

Hybrid EMG-IMU systems combine muscle activation sensing with movement and orientation sensing. EMG provides information about neuromuscular intent, while IMUs provide kinematic context such as arm orientation, motion, and posture. In principle, combining these modalities can improve gesture recognition, control robustness, and interpretation of user movement compared with using either modality alone.

5.1 Prior Hybrid Architectures

Researchers have combined EMG and IMU sensors in wearable systems for gesture recognition, teleoperation, and assistive control. Wu et al. developed a wearable American Sign Language recognition system using forearm surface EMG and wrist-mounted IMU sensing. In that system, EMG helped distinguish muscle activation patterns while the IMU provided motion and orientation information, producing higher recognition performance than single-sensor approaches [12].

Commercial wearable devices such as the Myo armband have also been used as hybrid EMG-IMU platforms. The Myo armband combines multiple EMG channels with inertial sensing,

allowing researchers to investigate gesture recognition, robotic control, and human-computer interaction using both muscle and motion data [8]. These systems demonstrate the value of combining physiological and kinematic information when richer control input is required.

For prosthetic and rehabilitation applications, hybrid sensing is attractive because EMG can indicate grasp or muscle intent while IMUs can provide information about limb position or movement. However, hybrid control is not automatically better for every application. It increases hardware requirements, calibration effort, data synchronisation complexity, and processing demand. For this reason, the present project focuses first on the simpler EMG acquisition stage before attempting hybrid fusion.

5.2 Challenges in Hybrid EMG-IMU Systems

A major challenge in hybrid systems is synchronisation. EMG and IMU signals may be sampled at different rates, filtered differently, and transmitted with different delays. If the two data streams are not aligned correctly, classification or control decisions may be based on mismatched muscle and movement information [8], [12].

Calibration is another major limitation. EMG performance depends on electrode placement, skin contact, muscle fatigue, and user-specific anatomy. IMU performance depends on sensor orientation, attachment stability, drift, and magnetometer interference. A hybrid system must therefore calibrate both the biological signal path and the motion-tracking path, increasing setup time and reducing ease of use [12].

System complexity also increases when multiple sensors are combined. Hybrid systems may require additional wiring, wireless communication, larger feature sets, more advanced classifiers, and greater processing power. This can be acceptable in laboratory research, but it may be less practical for low-cost rehabilitation or prosthetic systems where simplicity, reliability, and repeatability are critical [5], [8].

5.3 Actuation and Mechanical Constraints in Robotic Hands

Even when user intent is detected correctly, robotic hand control also depends on the mechanical and actuation system. A reliable EMG signal cannot produce useful prosthetic movement if the actuators, tendons, or joints are not repeatable. Robotic hands are affected by servo torque limits, backlash, response time, tendon slack, routing friction, and coupling between fingers.

Dexterous robotic hands may be fully actuated, where each joint or finger has more independent control, or underactuated, where one actuator drives multiple joints or fingers through mechanical coupling. Underactuated designs can reduce cost, weight, and control complexity, but they also reduce independent finger control. Fully actuated designs can provide greater dexterity, but they increase cost, weight, power demand, and mechanical complexity.

For low-cost prosthetic development, this creates a trade-off between affordability and precision. Low-cost actuators and tendon systems may be suitable for proof-of-concept demonstrations, but reliable prosthetic control requires sufficient torque, predictable movement, safe mechanical limits, and repeatable finger positioning. This supports the

present project’s decision to treat DexHand actuation as a future integration stage rather than as the main validated result.

5.4 Control Architecture Trends

Control architecture also affects responsiveness and reliability. Some systems use centralised processing on a PC, where sensor data are collected, analysed, and converted into robot commands externally. This can support complex algorithms and easier debugging, but it introduces dependence on the host computer and communication link.

Other systems use embedded or edge processing, where the microcontroller performs signal processing close to the sensor. This can reduce system complexity and support lower-latency control decisions, especially for simple tasks such as threshold-based intent detection. However, embedded systems have limited processing power, memory, and debugging flexibility compared with a full computer.

Communication method is another design factor. USB serial or UART communication is simple and reliable for laboratory logging, while Bluetooth or wireless links improve wearability but may introduce variable latency and connection issues [8]. More advanced robotic systems may use frameworks such as ROS or external controllers, but these increase integration complexity.

The present project adopts a simple embedded-first architecture: the Nano ESP32 samples and processes the EMG signal, while USB serial logging is used for capture and analysis. This approach matches the project aim of developing a low-cost, understandable, and measurable EMG acquisition stage before progressing toward more complex hybrid sensing or robotic actuation.

6. Comparative Analysis

The reviewed interface modalities differ in what they measure, how easily they can be implemented, and how suitable they are for rehabilitation-oriented robotic control. Vision systems observe external movement, EMG systems measure muscle activation, IMUs measure motion and orientation, and hybrid EMG-IMU systems combine physiological and kinematic information. Table 2.1 compares these approaches using the main criteria relevant to this project: accuracy, latency, ease of use, robustness, and rehabilitation value.

Table 2.1. Comparison of human-machine interface modalities for assistive robotic control.

Method	Main Strength	Main Limitation	Ease of Use	Rehabilitation/Prosthetic Relevance
Vision Based	Device-free observation of gestures, posture, and object interaction	Sensitive to lighting, occlusion, camera viewpoint, and visual noise	High for user comfort, but environment-dependent	Useful for gross movement tracking and teleoperation, less suitable for direct muscle-intent detection [13], [14], [15]
EMG Based	Direct measurement of muscle	Sensitive to electrode placement, skin	Moderate, because electrodes and	Highly relevant to myoelectric prosthetic control and grasp-intent detection [4], [7], [9]

	activation and user intent	contact, fatigue, and noise	calibration are required	
IMU Based	Lightweight tracking of limb motion and orientation	Drift, sensor misalignment, and magnetic interference can reduce accuracy	Moderate to high, depending on mounting and calibration	Useful for arm motion tracking, teleoperation, and rehabilitation exercise monitoring [10], [11], [17], [18]
Hybrid EMG-IMU	Combines muscle intent with kinematic context	Increased hardware, synchronisation, calibration, and processing complexity	Lower than single-sensor systems due to setup burden	Promising for richer wearable control, but more complex than the EMG-first approach used in this project [8], [12]

6.1 Suitability of Each Modality

Vision-based systems are strongest when non-contact interaction is required. They allow users to control or interact with robotic systems without wearing sensors, which makes them attractive for gesture control, teleoperation, and environmental awareness. However, vision depends strongly on lighting, viewpoint, occlusion, and camera quality. This makes it less reliable for detecting subtle muscle intent or fine grasp initiation.

EMG-based systems are strongest when the goal is to detect user muscle intent. EMG provides information about neuromuscular activation and is therefore highly relevant to myoelectric prostheses and assistive hand control. Its main disadvantages are electrode placement sensitivity, signal noise, fatigue effects, and the need for calibration. These issues directly relate to the present project, where signal repeatability and threshold reliability were key limitations.

IMU-based systems are strongest when continuous motion tracking is required. They are useful for estimating limb orientation, tracking exercises, and mapping human arm movement to robotic systems. However, IMUs do not directly measure muscle effort or grasp intent. They are therefore better suited to arm-position tracking than to detecting the muscle activation needed for prosthetic hand control.

Hybrid EMG-IMU systems offer the richest information because they combine physiological and kinematic sensing. EMG can indicate intent, while IMUs can describe limb movement and orientation. However, hybrid systems require more hardware, more calibration, and more complex data synchronisation. For this reason, hybrid sensing is best treated as a future extension rather than the first validated stage of a low-cost prosthetic control system.

6.2 Design Trade-Offs

The comparison shows that no single interface modality is ideal for every assistive robotic task. Vision systems are easy for the user but depend on the environment. IMUs are portable and useful for motion tracking but do not detect muscle intent. Hybrid systems provide richer information but increase system complexity.

EMG provides the most direct link to myoelectric prosthetic control, but it is also highly dependent on the quality of the skin-electrode interface. This creates a central trade-off for low-cost prosthetic research: the system must be affordable and simple enough to use, while

still producing a reliable signal. The present project therefore focuses on the most fundamental stage of the control chain: acquiring a usable EMG signal, processing it on accessible hardware, and evaluating whether it can produce a binary intent output.

This EMG-first approach does not reject vision, IMU, or hybrid control. Instead, it recognises that more advanced sensing and actuation should be built on top of a stable and repeatable muscle-signal acquisition pathway.

7. Identified Research Gap

The reviewed literature shows that vision, EMG, IMU, BCI, and hybrid sensing systems can each contribute to assistive robotic control. However, many advanced systems focus on high-level gesture recognition, teleoperation, or multi-sensor fusion, while the practical reliability of the initial EMG acquisition stage remains a major barrier for low-cost prosthetic development.

For myoelectric hand control, stable EMG acquisition is a foundational requirement. Before a system can support gesture classification, proportional grip control, or dexterous robotic actuation, it must first acquire a repeatable muscle signal and distinguish deliberate activation from rest. The literature shows that EMG can be effective for prosthetic and assistive control, but it is also sensitive to electrode placement, skin contact, fatigue, motion artefacts, and calibration conditions [4], [7], [9], [23].

A further gap exists between laboratory performance and practical low-cost implementation. Many systems report classification accuracy or demonstrate robotic movement, but fewer focus on the simple engineering questions that determine whether a low-cost EMG pathway is dependable: whether the signal can be logged consistently, whether a threshold can detect contraction without false triggers, whether repeated flex windows are detected reliably, and which setup factors reduce repeatability.

Hybrid EMG-IMU systems offer useful future potential by combining muscle intent with movement context [8], [12]. However, adding IMU sensing also increases hardware complexity, synchronisation requirements, and calibration burden. For an accessible prosthetic-hand platform, it is therefore important to first validate the simpler EMG sensing and intent-detection stage before progressing toward hybrid fusion or full DexHand actuation.

This project addresses this gap by developing and evaluating a low-cost EMG acquisition and binary intent-detection workflow using a MyoWare 2.0 sensor and Arduino Nano ESP32. The work focuses on signal acquisition, filtering, thresholding, serial logging, missed detections, false triggers, and practical setup limitations. In doing so, it provides a measured foundation for future prosthetic hand control without overstating the system as a fully validated robotic prosthesis.

8. Summary

The literature shows that several human-machine interface technologies can support assistive robotic control. Vision-based systems allow device-free gesture and movement tracking, but they are sensitive to lighting, viewpoint, and occlusion. IMU-based systems provide wearable motion and orientation tracking, but they can be affected by drift, magnetic interference, sensor placement, and calibration. BCI systems offer a potential pathway for users with limited motor function, but they remain complex and difficult to deploy reliably outside controlled environments.

EMG-based systems are the most directly relevant to myoelectric prosthetic control because they measure muscle activation and can provide an indication of user intent. However, the literature also shows that EMG signals are sensitive to electrode placement, skin contact, fatigue, motion artefacts, and calibration. These limitations are especially important for low-cost prosthetic development, where the system must remain simple, repeatable, and accessible.

Hybrid EMG-IMU systems offer a promising future direction by combining muscle intent with movement context. However, hybrid sensing also introduces additional hardware, synchronisation, processing, and calibration requirements. For this reason, a reliable EMG acquisition stage is an important prerequisite before more complex multi-sensor control is attempted.

This review therefore supports the direction of the present project: developing and evaluating a low-cost EMG acquisition and binary intent-detection workflow as a foundation for future robotic hand control. The project focuses on the practical first stage of the control chain: acquiring a usable forearm EMG signal, processing it on accessible embedded hardware, logging it for analysis, and assessing whether deliberate contraction can be distinguished from rest.

Chapter 3 — Methodology

3.1 Methodological Overview

This chapter describes the methodology used to develop and evaluate a low-cost surface electromyography (sEMG) acquisition and intent-detection system for future prosthetic or assistive robotic hand control. The original project concept considered a broader hybrid sensing architecture involving both electromyography (EMG) and inertial measurement unit (IMU) sensing. However, the final validated methodology was narrowed to the EMG signal path in order to produce a complete, measurable, and defensible experimental system within the available project timeframe.

The central methodological aim was to determine whether a low-cost wearable EMG sensor and embedded microcontroller could acquire forearm muscle activation, process the signal in real time, and convert the result into a binary intent output suitable for future robotic hand control. The project therefore focused on the acquisition and interpretation stage that would sit upstream of any prosthetic actuation system. A 3D-printed DexHand platform was retained as the intended application context, but full hand actuation was not treated as a requirement for validating the EMG sensing pipeline.

The validated system chain was defined as:

forearm muscle activation -> MyoWare EMG sensor -> Arduino Nano ESP32 -> filtering and thresholding -> intent signal

This approach allowed the work to be evaluated using practical engineering measures such as signal separation, repeatability, missed detections, false triggers, and response behaviour during repeated contraction trials.

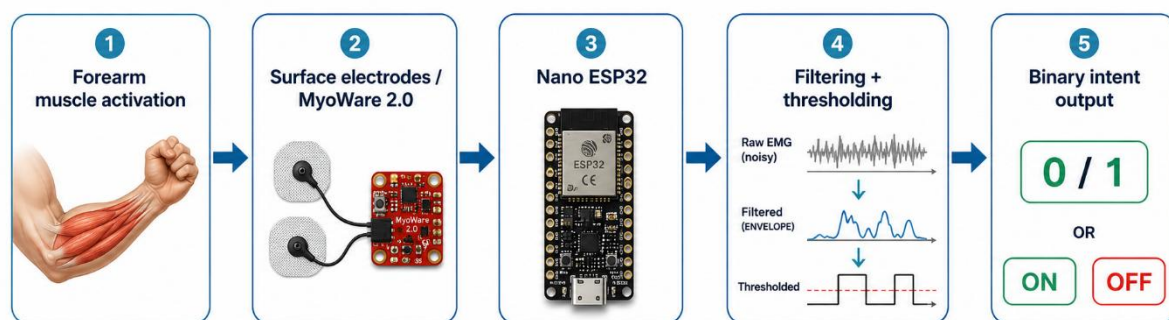


Figure 3.1. Validated EMG acquisition and intent-detection pathway used in the project.

3.1.1 System Boundaries and Functional Blocks

The system was conceptualised as a human-machine signal acquisition chain consisting of four main functional blocks.

1. Human neuromuscular input

The user's forearm muscle activation acted as the biological input source. When the participant voluntarily contracted the target forearm muscle group, electrical activity generated by muscle fibres was detected at the skin surface using disposable surface electrodes. The human body was treated as an external input source rather than part of the designed electronic system.

2. EMG sensing layer

The sensing layer consisted of the SparkFun MyoWare 2.0 muscle sensor ecosystem and surface electrodes. This layer converted muscle activity into an analog output voltage that could be read by a microcontroller. The envelope-style output was selected for the main experimental path because it provided a more immediately usable activation signal than a raw EMG waveform.

3. Processing and logging layer

An Arduino Nano ESP32 was used as the embedded processing platform. It sampled the MyoWare output using its analog input, converted the reading into voltage, estimated a baseline, extracted a positive activation feature, applied smoothing, and compared the resulting signal against an adaptive threshold. The processed stream was transmitted to a host computer over USB serial and logged as CSV data for analysis.

4. Intended actuation/output layer

The output of the validated system was a binary intent state representing rest or detected contraction. This intent signal was intended to provide the basis for later robotic hand actuation. The DexHand platform and servo driver hardware were therefore treated as target output hardware rather than as the primary source of experimental validation.

The system boundary for the validated thesis work included the sensing layer, processing layer, data logging, and analysis pipeline. The robotic hand was included as an application context and mechanical prototype, but the core experimental claim was based on EMG acquisition and processing.

3.1.2 EMG Signal Pathway

The implemented methodology used a single EMG pathway rather than a full hybrid EMG-IMU pathway. The signal path was:

muscle contraction -> surface electrodes -> MyoWare 2.0 -> Nano ESP32 A0 -> filtering -> threshold decision -> intent output

The pathway was designed to detect a simple control intention: the difference between relaxed forearm state and deliberate contraction. This binary approach was selected because it is an appropriate first step for low-cost prosthetic control. Before a system can support multi-gesture recognition or proportional finger movement, it must first demonstrate that it can acquire a stable and repeatable muscle activation signal.

Although IMU sensing was considered during the wider concept stage, it was not included in the final validated experimental method. IMU sensing remains relevant for future arm orientation tracking, but including it in the final methodology would have increased integration complexity without directly improving the validation of the EMG acquisition chain.

3.1.3 Hardware Selection Methodology

Hardware was selected according to four practical criteria:

- low cost and accessibility
- compatibility with student-level prototyping
- ability to produce measurable EMG data
- suitability for future robotic hand integration

The MyoWare 2.0 muscle sensor was selected because it is a compact and accessible surface EMG module designed for wearable muscle-signal experimentation. It provides outputs that are suitable for microcontroller-based acquisition and allows rapid testing of electrode placement and muscle activation response.

The Arduino Nano ESP32 was selected as the primary embedded controller because it provides a compact development platform with USB serial communication, a 12-bit analog-to-digital converter, and sufficient processing capability for lightweight filtering and threshold logic. The Nano ESP32 also supports future expansion to wireless communication or additional peripherals if required.

The DexHand platform was selected as the mechanical target because it is an open-source, 3D-printable dexterous hand concept. This made it appropriate for a low-cost prosthetic research project. However, the methodology did not depend on achieving full DexHand actuation, because the central research contribution was the EMG acquisition and interpretation stage.

3.2 Hardware Design and Selection

3.2.1 EMG Sensing Hardware

The EMG sensing system used a SparkFun MyoWare 2.0 muscle sensor development kit with disposable surface electrodes. A single-channel acquisition configuration was used, targeting the forearm flexor muscle region. A single channel was sufficient for the validated binary intent-detection task, which aimed to distinguish between rest and deliberate contraction.

The electrode cable was connected to the MyoWare Cable Shield, which was snapped onto the MyoWare Muscle Sensor. The sensor output was connected to the Arduino Nano ESP32 using three main connections:

- VIN -> Nano ESP32 3V3

- GND -> Nano ESP32 GND
- ENV -> Nano ESP32 A0

The ENV output was selected as the main input because it provided an envelope-style representation of muscle activation. This reduced the immediate need for complex analog signal conditioning and made the acquisition path more practical for a time-limited undergraduate prototype. The sensor was powered from the Nano ESP32 3.3 V rail so that the analog output remained within the safe input range of the microcontroller.



Figure 3.2. EMG acquisition hardware setup showing the MyoWare 2.0 sensor connection, Arduino Nano ESP32, electrode cable, and USB serial link used for data capture.

3.2.2 Electrode Type and Placement

Disposable surface electrodes were used because they are low cost, widely available, and appropriate for short-duration EMG testing. The target muscle group was the forearm flexor

region because it is associated with wrist and finger flexion and provides a practical proxy for grasp intent.

Electrode placement was carried out using a practical palpation method. The participant placed the forearm in a supported palm-up position and performed a firm wrist flexion or fist contraction. The active electrodes were placed along the direction of the muscle fibres over the region that became firm during contraction. The reference electrode was placed on a quieter area away from the active muscle belly, typically near a bony region of the wrist or side of the forearm.

The practical placement rule used during testing was:

- active electrode 1 over the forearm muscle belly
- active electrode 2 approximately 2-4 cm away, aligned with the muscle direction
- reference electrode on a quieter region away from the primary contraction



Figure 3.3. Surface electrode placement on the forearm for single-channel EMG acquisition, showing active electrode positioning and the reference electrode location used during testing.

3.2.3 Processing Platform

The Arduino Nano ESP32 acted as the embedded acquisition and processing unit. It sampled the MyoWare ENV signal on analog input A0 using a 12-bit ADC configuration. The firmware used a sampling interval of approximately 4 ms, corresponding to an estimated sample rate of 250 Hz.

The Nano ESP32 transmitted data to a host computer using USB serial communication. This enabled live testing in the Arduino Serial Monitor and structured data capture using a custom PowerShell logging script. USB serial was selected because it was reliable, simple to debug, and suitable for collecting timestamped trial data.

The processing platform was responsible for:

- analog sampling
- voltage conversion
- baseline estimation
- activation feature extraction
- digital smoothing
- adaptive thresholding
- binary intent output
- CSV-format serial transmission

3.2.4 Robotic Hand Platform

The DexHand platform was used as the intended future robotic target for the project. It provided a physical representation of the type of dexterous hand that could eventually be controlled using the EMG intent signal. The use of an open-source, 3D-printable hand aligned with the project theme of low-cost prosthetic and assistive technology.

The hand was treated as an experimental platform and demonstration prototype only. It was not presented as a medical device and was not clinically evaluated. The methodology therefore separated the validated EMG acquisition work from the future mechanical actuation work.

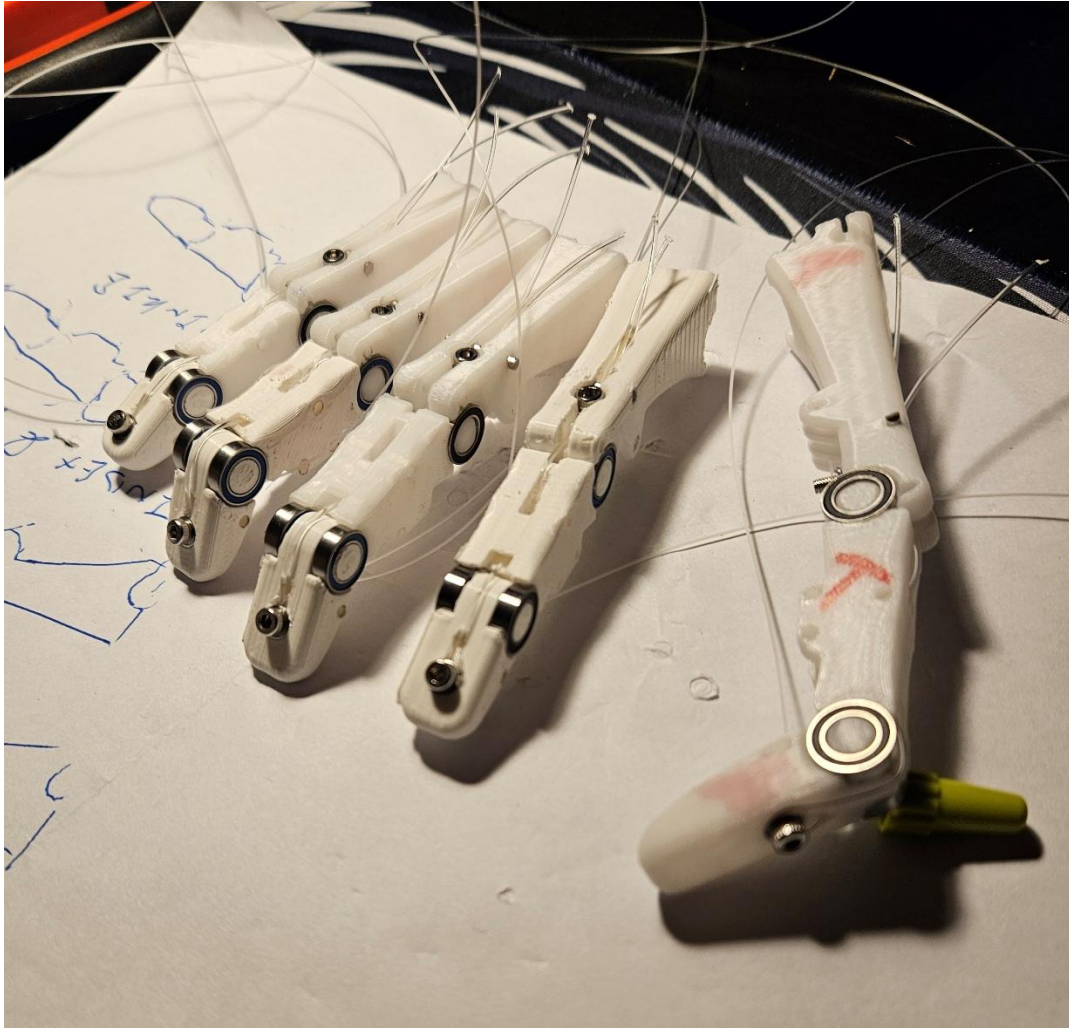


Figure 3.4. 3D-printed DexHand prototype used as the intended future mechanical target for EMG-driven actuation.

3.2.5 Deferred IMU and Robotic Arm Integration

IMU sensing and robotic arm integration were considered during the concept stage because they could support future upper-limb orientation tracking and arm teleoperation. However, these elements were not included in the final validated methodology. This decision was made to avoid overextending the project and to prioritise a complete EMG acquisition and processing pipeline.

In future work, an IMU could be added to measure wrist or forearm orientation, while the EMG signal could remain responsible for grasp intent. This would create a complementary sensing architecture in which EMG detects intent and IMU sensing provides motion context. In the present thesis, however, IMU and robotic arm control are treated as future extensions rather than implemented results.

3.3 EMG Signal Processing Pipeline

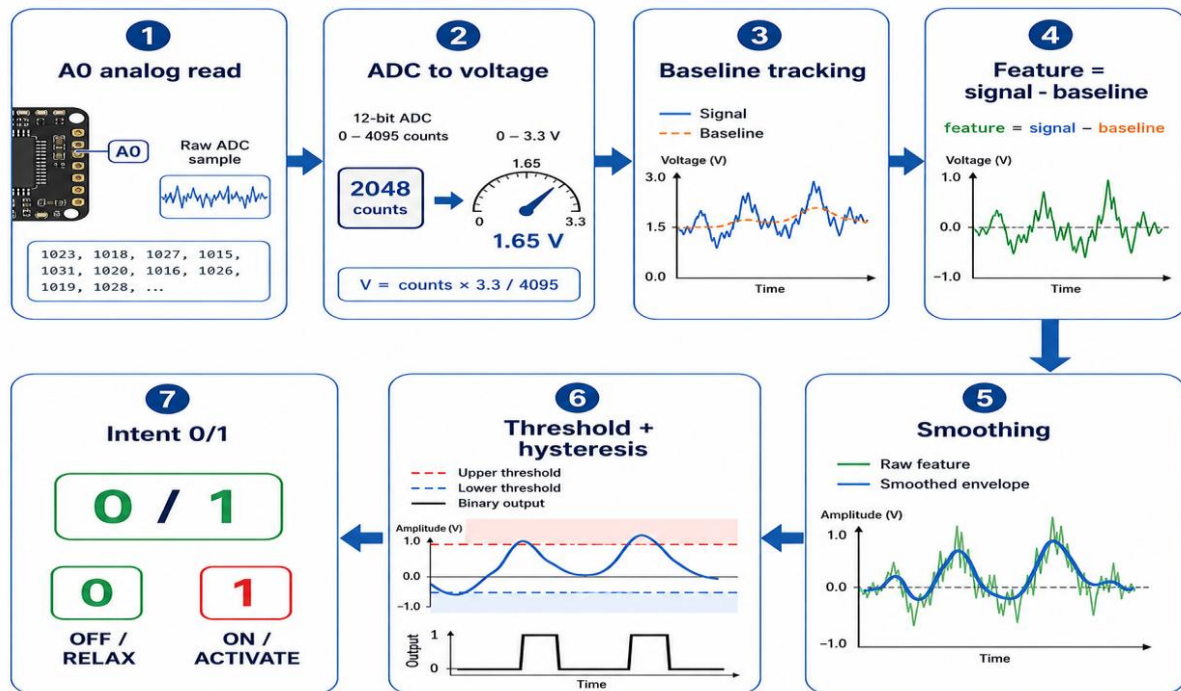


Figure 3.5. EMG processing pipeline implemented on the Nano ESP32, showing analog acquisition, voltage conversion, baseline tracking, feature extraction, smoothing, thresholding, and binary intent output.

3.3.1 Signal Acquisition

The Nano ESP32 acquired the MyoWare ENV analog output at approximately 250 Hz. Each sample was read using the microcontroller analog input and converted from ADC count to voltage using:

$$V = (\text{ADC_count} / 4095) * 3.3$$

where 4095 is the maximum value of the 12-bit ADC and 3.3 V is the reference voltage used for the signal path.

The firmware transmitted each sample as a CSV row containing:

- sample timestamp in milliseconds
- raw ADC value
- voltage
- baseline estimate
- extracted activation feature

- filtered activation value
- threshold value
- noise estimate
- binary intent state
- state-change flag
- protocol marker

This structure allowed both real-time observation and offline analysis. The CSV format also made it possible to compare the recorded EMG signal against the rest/flex protocol markers during post-processing.

3.3.2 Baseline Tracking

The EMG output was not treated as a fixed absolute value because surface EMG readings vary with electrode contact, skin impedance, cable movement, and posture. A slowly adapting baseline was therefore used to estimate the current resting level of the signal.

The baseline was updated using exponential smoothing:

$$\text{baseline}[n] = \text{baseline}[n-1] + \alpha * (\text{signal}[n] - \text{baseline}[n-1])$$

A faster baseline update was used when the system was idle, while a slower update was used when contraction intent was active. This prevented the baseline from rapidly adapting upward during a sustained contraction and reducing the apparent activation.

3.3.3 Activation Feature Extraction

The MyoWare envelope output was already partially processed by the sensor hardware. The firmware therefore extracted a positive activation feature by subtracting the estimated baseline from the measured voltage and clamping negative values to zero:

$$\text{feature}[n] = \max(\text{signal}[n] - \text{baseline}[n], 0)$$

This feature represented how far the current signal rose above the estimated resting level. It was more suitable for threshold-based intent detection than the raw ADC value because it reduced dependence on the absolute resting offset.

3.3.4 Digital Filtering

The activation feature was smoothed using two exponential filters. A faster smoothing stage responded quickly to rising activation, while a slower smoothing stage helped reduce jitter and preserve the broader envelope of the contraction. The filtered output used the greater of the fast and slow smoothed values.

The smoothing equation was:

$$\text{filtered}[n] = \text{filtered}[n-1] + \alpha * (\text{feature}[n] - \text{filtered}[n-1])$$

This approach was selected because it is computationally lightweight and suitable for microcontroller implementation. It does not require large buffers or complex frequency-domain processing. This was important because the aim was to develop a practical embedded acquisition pipeline rather than an offline-only signal processing method.

3.3.5 Thresholding and Intent Detection

Binary intent detection was implemented using adaptive thresholding and hysteresis. The threshold was calculated using an estimated noise floor, a scaling factor, and a minimum threshold value:

$$\text{threshold} = \max(\text{minimum_threshold}, \text{noise_floor} * \text{scale} + \text{offset})$$

When the filtered activation exceeded the threshold, the system switched to an active intent state. The state was only released once the filtered signal fell below a lower hysteresis level. This prevented rapid switching when the signal hovered near the threshold.

The resulting output was binary:

- 0 = rest or no detected contraction
- 1 = detected muscle activation

This binary method was selected because it matched the validated experimental scope and provided a realistic first stage for future prosthetic hand control. More advanced control methods, such as multi-gesture classification or proportional grip control, would require stronger signal repeatability and additional validation.

3.4 Experimental Protocol

A guided repeated-contraction protocol was used to create a repeatable basis for EMG evaluation. Each structured capture used the RepeatedFlex5 protocol, consisting of:

- 7 s settling period
- five repeated cycles, each consisting of 3 s rest and 3 s flex

During the settling period, the participant kept the forearm relaxed so that the system could stabilise and estimate an initial baseline. During each rest window, the participant relaxed the forearm as much as possible. During each flex window, the participant performed a deliberate wrist/finger flexion or fist contraction to activate the target forearm muscle group.

The protocol was controlled using a host-side PowerShell capture script. The script displayed rest and flex cues and sent marker commands to the Nano ESP32 at each transition. These markers were saved alongside the EMG sample stream so that the recorded data could be compared against known rest and contraction windows during offline analysis.

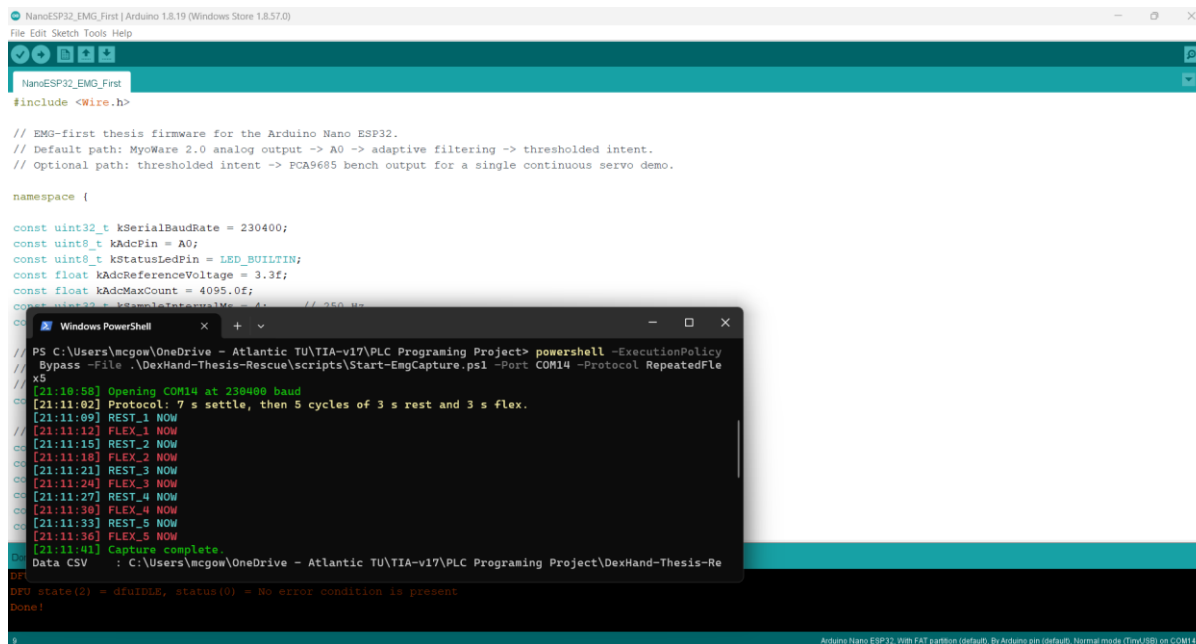


Figure 3.6. *RepeatedFlex5* experimental protocol used for EMG capture, consisting of a 7 s settling period followed by five repeated 3 s rest and 3 s flex windows.

The protocol was designed to evaluate whether deliberate contraction could be distinguished from rest using the recorded EMG signal. It also allowed missed detections, false triggers, and onset response behaviour to be assessed in a structured way.

3.5 Performance Metrics

The system was evaluated using practical engineering metrics suited to a single-channel binary EMG intent-detection task. These metrics were selected to assess whether the system could distinguish deliberate contraction from rest and whether the thresholding logic behaved safely during repeated trials.

3.5.1 Signal Separation

Signal separation was assessed by comparing the recorded resting signal against the recorded contraction signal during the guided rest/flex protocol. A useful EMG response was defined as one where deliberate contraction produced a measurable increase above the resting signal band. This metric was important because reliable intent detection depends first on the presence of a distinguishable signal change.

3.5.2 Missed Detections

A missed detection occurred when an instructed flex window did not contain a threshold crossing. This metric indicated whether the filtering and thresholding pipeline was sensitive enough to detect deliberate contraction. A high number of missed detections suggested that the contraction signal was too weak, the threshold was too conservative, or the electrode setup did not provide sufficient signal margin.

3.5.3 False Triggers

A false trigger occurred when the intent state became active outside an instructed flex window. This metric was important because an assistive or prosthetic hand controller should not move unintentionally during rest. For safety and usability, avoiding false triggers was treated as especially important, even if this made the threshold more conservative.

3.5.4 Onset Response

Onset response was estimated by comparing the timestamp of a flex cue marker with the first detected threshold crossing inside that flex window. This measurement does not represent clinical neuromuscular latency. Instead, it represents the response of the implemented sensor-processing pipeline under the test conditions used in this project.

3.5.5 Repeatability

Repeatability was assessed by comparing behaviour across the five repeated rest/flex cycles. A repeatable system should show similar signal increases during flex windows and should return toward baseline during rest windows. Repeatability was evaluated using both the plotted signal behaviour and the summary statistics generated from the recorded session data.

3.6 Methodological Limitations

Several limitations were recognised in the methodology. First, the test participant was the project author, so the results should be interpreted as a single-participant feasibility study rather than a clinical validation. No amputee users, prosthesis users, or neurologically impaired participants were included.

Second, the system used one EMG channel and a binary intent output. It therefore did not validate multi-channel gesture recognition, proportional finger control, independent finger movement, or continuous prosthetic control. The binary output was treated as an early-stage intent signal rather than a complete hand-control solution.

Third, surface EMG proved highly sensitive to electrode placement, skin preparation, cable strain, contact pressure, and posture. These factors affected signal quality and repeatability. Rather than being ignored, these practical issues were treated as part of the engineering evaluation because they directly affect the feasibility of low-cost EMG control.

Finally, although the DexHand platform was included as the intended application target, the thesis methodology did not depend on full actuation of the hand. This allowed the work to focus on the most defensible technical contribution: acquisition, logging, filtering, and interpretation of a low-cost EMG control signal.

These limitations define the scope of the methodology and help separate the validated EMG acquisition work from future prosthetic actuation, multi-channel sensing, and clinical testing.

Chapter 4 — System Design and Implementation

4.1 Overall System Architecture

The implemented system was designed as a low-cost EMG acquisition and intent-detection platform for future robotic hand control. The final implementation focused on the sensing, processing, logging, and analysis chain rather than full prosthetic actuation. This decision allowed the project to produce a measurable and defensible system while still retaining the DexHand as the intended future mechanical target.

The final validated architecture was:

surface electrodes -> MyoWare 2.0 muscle sensor -> Arduino Nano ESP32 filtering and thresholding -> serial logging -> analysis and visualisation

The intended future control architecture is:

surface electrodes -> MyoWare 2.0 -> Arduino Nano ESP32 -> filtered intent signal -> servo driver -> DexHand movement

The difference between these two architectures is important. The first represents the system implemented, tested, and analysed in this project. The second represents the future integration pathway required for prosthetic hand actuation. This distinction prevents the project from overstating the validated results while still showing how the EMG intent signal could be used in later DexHand control.

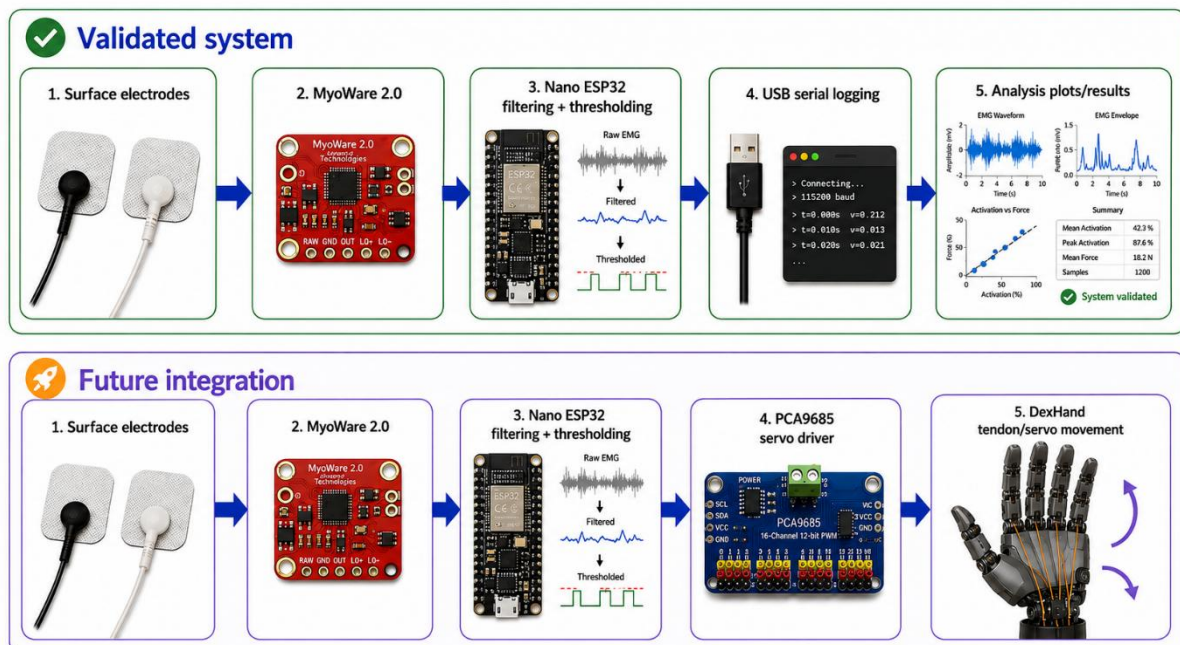


Figure 4.1. Overall system architecture showing the validated EMG acquisition and analysis pathway, alongside the intended future pathway for servo-driven DexHand actuation.

4.2 EMG Hardware Implementation

4.2.1 MyoWare Sensor Connection

The MyoWare 2.0 muscle sensor was connected to the Arduino Nano ESP32 using the VIN, GND, and ENV connections. The ENV signal was selected because it provided a processed envelope-style output that could be read directly by the microcontroller analog input. This simplified the implementation and made the system more practical for rapid testing and embedded thresholding.

The final wiring used was:

- MyoWare VIN -> Nano ESP32 3V3
- MyoWare GND -> Nano ESP32 GND
- MyoWare ENV -> Nano ESP32 A0

The sensor was powered from the Nano ESP32 3.3 V rail rather than a higher-voltage supply so that the analog output would remain within the safe input range of the microcontroller. This reduced the risk of damaging the analog input during testing.

During development, unstable jumper connections caused unreliable readings and intermittent signal loss. To improve mechanical reliability, header pins were soldered to the MyoWare output connections so that the VIN, GND, and ENV jumper wires could be connected more securely. This was an important practical improvement because unstable electrical contact can appear as signal noise, drift, or false biological behaviour.

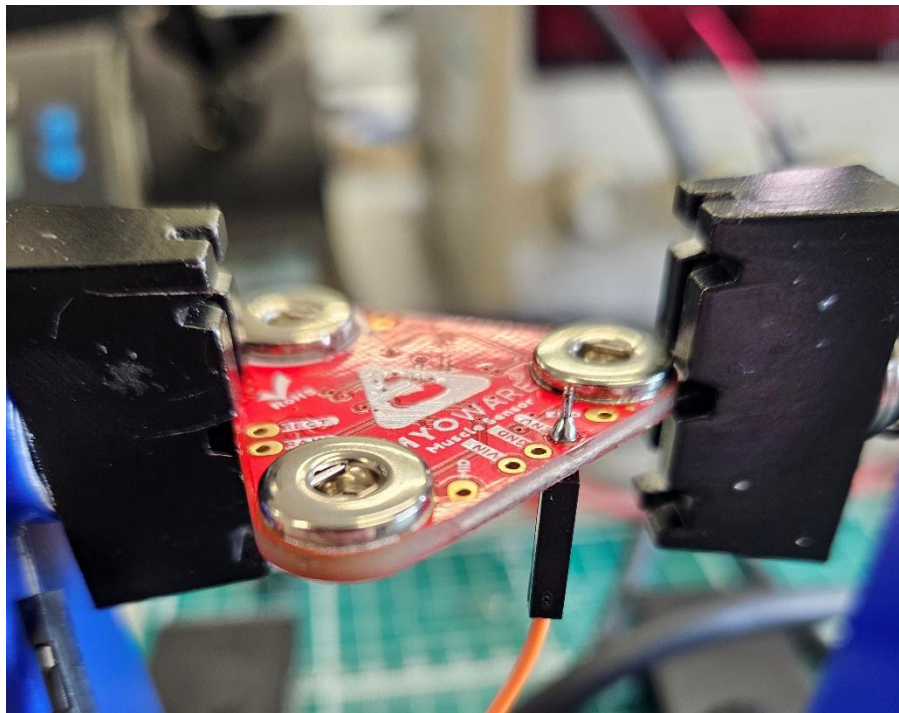


Figure 4.2. MyoWare 2.0 to Arduino Nano ESP32 wiring arrangement, showing the VIN, GND, and ENV signal connections used for EMG acquisition.

4.2.2 Electrode Cable and Skin Interface

The electrode cable was connected to the MyoWare Cable Shield, which was snapped onto the MyoWare Muscle Sensor. Disposable surface electrodes were attached to the forearm. The active electrodes were placed along the target muscle direction and the reference electrode was placed on a quieter area near the wrist or side of the forearm.

In practice, this skin-electrode interface was one of the most sensitive parts of the system. Signal quality changed significantly with electrode placement, skin preparation, contact pressure, and cable movement. Fresh electrodes and skin cleaning improved consistency, while reused pads or cable strain could produce drifting values or weak contraction response.

The electrode setup therefore required:

- clean and dry skin
- fresh or strongly adhesive electrode pads
- active electrodes aligned with the muscle direction
- reference electrode placed away from the active muscle belly
- cable strain relief so the pads were not pulled during movement

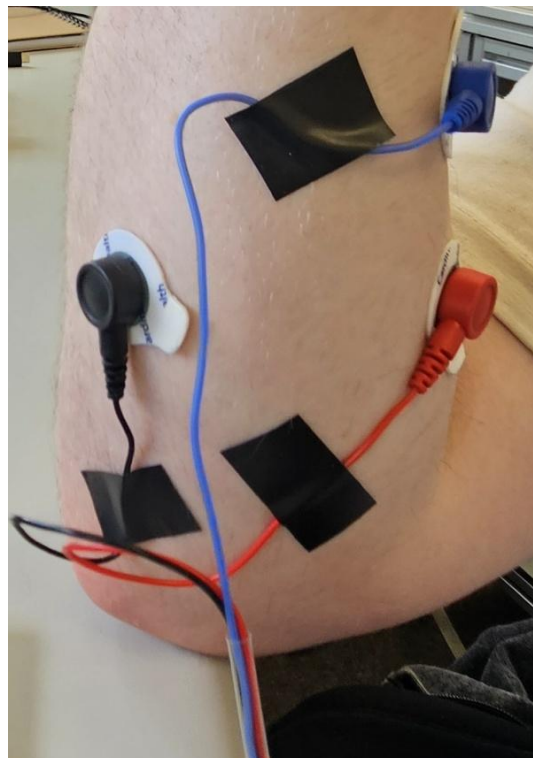


Figure 4.3. Forearm electrode placement and cable strain relief used during EMG testing, showing the practical skin-contact setup required for repeatable signal acquisition.

This practical sensitivity is typical of low-cost surface EMG acquisition and became an important limitation discussed later in the thesis.

4.3 Embedded Firmware Implementation

The Nano ESP32 firmware was written to acquire, process, threshold, and stream EMG data in real time. The firmware sampled the MyoWare ENV output on analog input A0 at approximately 250 Hz using a 4 ms sample interval. Each sample was converted from raw ADC count to voltage and then passed through the embedded processing pipeline.

The implemented firmware flow was:

analog read -> voltage conversion -> baseline tracking -> activation feature -> smoothing -> threshold comparison -> intent state -> serial output

The firmware output one CSV row per sample. Each row contained:

- sample timestamp
- raw ADC count
- voltage
- estimated baseline voltage
- activation feature voltage
- filtered voltage
- threshold voltage
- noise-floor estimate
- binary intent state
- state-change flag
- protocol marker

The binary intent output was used as the main control result. It represented whether the filtered activation signal had crossed the current threshold. A value of 0 represented rest or no detected contraction, while a value of 1 represented detected muscle activation.

This firmware design was intentionally lightweight so that the signal processing could run directly on the microcontroller. This supported the project aim of developing a low-cost embedded EMG acquisition system rather than relying only on offline processing.

4.4 Host Logging and Analysis Implementation

USB serial communication was used to transmit processed EMG data from the Nano ESP32 to a host computer. A PowerShell capture script was developed to log the serial stream to a CSV file. The script also supported the guided RepeatedFlex5 protocol by sending marker commands to the Nano ESP32 at specific rest and flex transitions.

The host logging system generated three main files for each capture:

- data CSV containing the EMG sample stream
- marker CSV containing the timing of rest/flex cues
- serial log containing configuration and diagnostic messages

A separate analysis script imported the captured CSV and marker files. It produced:

- raw/feature plot
- filtered signal and threshold plot
- latency or response table
- summary CSV
- Markdown analysis report

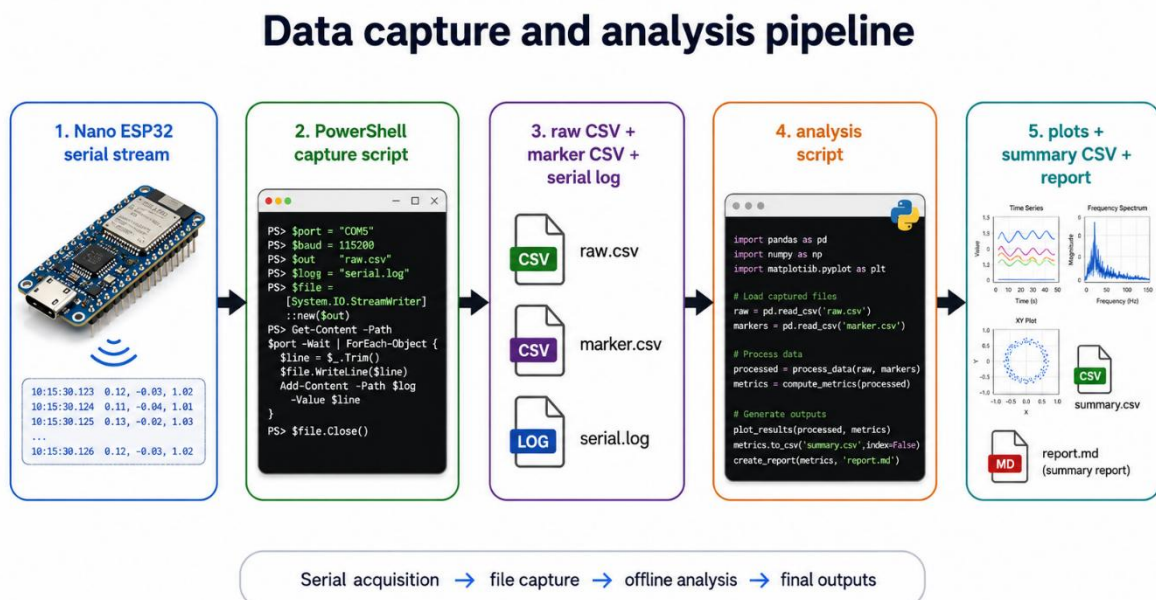


Figure 4.4. Host-side data capture and analysis pipeline showing how the Nano ESP32 serial stream was logged using a PowerShell capture script, stored as CSV and marker files, and processed into plots, summary metrics, and an analysis report.

This workflow separated the real-time embedded system from the offline analysis stage. The Nano ESP32 performed lightweight acquisition, filtering, and thresholding, while the host computer handled structured data capture, plotting, and reporting. This made the system easier to debug and allowed the same recorded session to be reviewed after testing.

4.5 Mechanical System Design

4.5.1 DexHand Platform

The mechanical target for the project was an open-source 3D-printed dexterous hand platform based on the DexHand design. The printed hand provided a physical representation of the type of robotic prosthetic system that could be controlled using the EMG intent signal in future work.

The DexHand platform was appropriate for this project because:

- it aligned with the low-cost and open-source theme
- it could be produced using accessible 3D printing methods
- it represented a realistic target for future servo actuation
- it provided a visible and engaging prototype for the project demonstration

The printed hand was treated as an experimental robotic platform, not as a certified prosthetic device. Its role in this thesis was to provide an application target and mechanical proof-of-concept context. The validated experimental contribution remained the EMG acquisition and intent-detection system rather than full mechanical hand control.



Figure 4.5. Assembled 3D-printed DexHand finger segment showing joint hardware and fishing-line tendon routing. The prototype was used to explore the future mechanical target for EMG-driven actuation, while full hand actuation remained outside the validated scope of this project.

4.5.2 Tendon Routing and Fishing Line Limitation

The DexHand design uses tendon-like transmission to pull finger joints through routed lines. Fishing line was considered as a low-cost tendon material because it is inexpensive, available, flexible, and easy to thread through small printed parts. However, testing and assembly indicated that fishing line can be difficult to tension consistently and may reduce repeatability.

The main limitations of fishing line were:

- inconsistent tension between fingers
- tendency to slacken during handling
- difficulty anchoring securely
- sensitivity to routing friction
- limited repeatability for precise motion

This does not mean that low-cost tendon systems are unsuitable. Rather, it shows that low-cost design must still account for reliability and control precision. A future version should investigate improved tendon materials, better anchoring methods, smoother routing paths, and mechanical tension adjustment.

This lesson is important because the goal is not simply to make the cheapest hand possible. The goal is to identify the lowest-cost design that remains reliable, controllable, and useful.

4.6 Actuation System Considerations

4.6.1 Servo Driver Hardware

The project hardware included PCA9685-based 16-channel PWM servo driver boards. These boards are useful for future robotic hand control because they allow multiple servos to be controlled over an I2C connection while reducing the number of microcontroller pins required for PWM output.

In a future integrated system, the intended wiring would be:

- Nano ESP32 SDA/SCL -> PCA9685 I2C inputs
- external 5 to 6 V servo supply -> PCA9685 V+
- Nano ESP32 GND -> PCA9685 GND
- servo signal outputs -> individual servos

A separate servo power supply would be required because the Nano ESP32 cannot supply enough current for multiple servos directly. All grounds would need to be connected together so that the control signals share a common electrical reference.

4.6.2 Servo Selection Limitation

The available servos included 360-degree continuous-rotation micro servos. These are useful for demonstrating speed and direction control, but they are not appropriate for precise robotic finger positioning. A continuous-rotation servo does not move to a commanded angle in the same way as a positional servo. Instead, the control pulse usually affects rotation speed and direction.

For controlled DexHand finger movement, future work should use positional micro servos rather than continuous-rotation servos. Important selection criteria include:

- sufficient torque for tendon tension
- accurate position control
- low backlash
- reliable centring behaviour
- stable response to PWM commands
- availability of replacements
- suitable size for the printed hand structure

This limitation is important because low cost alone does not guarantee useful prosthetic control. Actuator quality directly affects reliability, precision, safety, and user confidence. The project therefore treated servo-driven DexHand actuation as a future integration task rather than as a validated experimental outcome.

4.7 Software Architecture

The implemented software architecture consisted of three layers.

1. Embedded acquisition firmware

The Nano ESP32 firmware acquired the analog EMG signal, performed baseline tracking, filtering, thresholding, and transmitted the processed data over USB serial.

2. Host capture script

The PowerShell capture script opened the serial port, ran the RepeatedFlex5 rest/flex protocol, sent marker commands to the Nano ESP32, and saved the sample stream to CSV files.

3. Host analysis script

The analysis script imported the captured data and marker files, calculated summary metrics, generated plots, and produced a Markdown analysis report.

This three-part architecture made the system easier to test and debug. Live signal checks could be performed in the Arduino Serial Monitor, structured protocol captures could be performed using PowerShell, and the resulting data could be analysed offline without changing the firmware. This separation also meant that the microcontroller only needed to run lightweight embedded processing, while the host computer handled plotting and reporting.

4.8 System Integration Challenges

Several integration challenges were encountered during implementation. These challenges were important because they affected the repeatability and reliability of the EMG acquisition workflow.

First, electrode placement and skin contact strongly affected signal quality. The same hardware could produce a strong response in one session and a weak or unstable response in another session depending on electrode position, cable tension, skin preparation, and electrode adhesion. This showed that the human-sensor interface was one of the most critical parts of the system.

Second, loose jumper wires caused unstable readings during early testing. This issue was addressed by soldering header pins to the MyoWare output connections and using more secure wiring. This improved mechanical connection reliability and reduced the likelihood of intermittent signal loss being mistaken for biological signal variation.

Third, the serial capture pipeline required careful port handling. The Arduino Serial Monitor had to be closed before running the PowerShell capture script, otherwise the COM port could be blocked. On the Nano ESP32, upload issues were sometimes resolved by entering bootloader mode using a double reset.

Fourth, the mechanical DexHand integration introduced practical issues relating to tendon tension, tendon routing, printed tolerances, and servo suitability. These challenges supported the decision to prioritise EMG acquisition and signal processing as the validated thesis contribution, while treating full DexHand actuation as future integration work.

4.9 Final Implemented System State

At the end of implementation, the project produced a working EMG acquisition and intent-detection workflow using the MyoWare 2.0 sensor and Arduino Nano ESP32. The system was capable of live analog signal reading, embedded filtering and thresholding, structured serial capture, offline analysis, and generation of plots and summary metrics.

The DexHand platform was developed as the intended mechanical target and project demonstration prototype, providing a clear application context for future prosthetic hand control. However, full DexHand actuation was not treated as the validated outcome of this project. Instead, the validated implementation focused on the sensing and processing pathway required before reliable actuation can be achieved.

The final system should therefore be understood as a low-cost EMG acquisition and binary intent-detection prototype with a clear pathway toward future servo-driven robotic hand control.

Chapter 5 — Experimental Setup and Results

5.1 Experimental Overview

This chapter presents the experimental setup and results for the EMG acquisition and intent-detection system. The purpose of testing was to evaluate whether the low-cost MyoWare 2.0 and Arduino Nano ESP32 signal chain could acquire forearm muscle activity, process the signal, and produce a usable binary intent output.

The experimental work focused on:

- live signal verification
- electrode placement trials
- structured rest/flex capture
- serial data logging
- offline signal analysis
- identification of practical limitations

Testing was conducted as a single-participant feasibility study, with the project author acting as the participant. The results should therefore be interpreted as bench-level engineering validation rather than clinical evidence. The chapter focuses on measured EMG acquisition and threshold-detection behaviour, not on clinical prosthetic performance or validated DexHand actuation.

5.2 Evidence Hierarchy and Claim Boundaries

Three levels of evidence were used during evaluation, and each level supports a different strength of claim.

First, live setup observations from the Arduino Serial Monitor were used to confirm that the sensor and microcontroller responded to forearm contraction under favourable conditions. These observations were useful during setup because they showed whether electrode placement and wiring were producing a visible signal response. However, live observations were not treated as the strongest formal evidence unless supported by saved data.

Second, structured CSV captures were used as the main experimental record. These files provided timestamped samples, protocol markers, and processed signal values that could be analysed after the session. A logged CSV therefore provides stronger evidence than a live observation because it can be plotted, measured, and reviewed.

Third, threshold-crossing results from the analysis pipeline were treated as evidence of binary intent detection. A successful detection required the filtered signal to cross the adaptive threshold during an instructed flex window. Therefore, the thesis distinguishes between three

outcomes: observing a live muscle response, recording a raw signal change in a CSV file, and successfully detecting intent through the threshold algorithm.

This distinction prevents overclaiming. The project can safely claim that the acquisition and logging workflow was implemented and that live EMG response was observed under favourable conditions. Successful binary intent detection is only claimed where logged data show threshold crossings during instructed flex windows.

5.3 Experimental Setup

The test setup consisted of the following components:

- SparkFun MyoWare 2.0 muscle sensor
- MyoWare Cable Shield and electrode cable
- disposable surface electrodes
- Arduino Nano ESP32
- USB serial connection to host computer
- PowerShell capture script
- analysis script for plots and summary metrics

The sensor output was connected to the Nano ESP32 using:

- VIN -> 3V3
- GND -> GND
- ENV -> A0

The participant placed the forearm in a supported position to reduce motion artefacts. The active electrodes were aligned with the forearm flexor muscle direction, and the reference electrode was placed on a quieter area away from the main muscle belly.

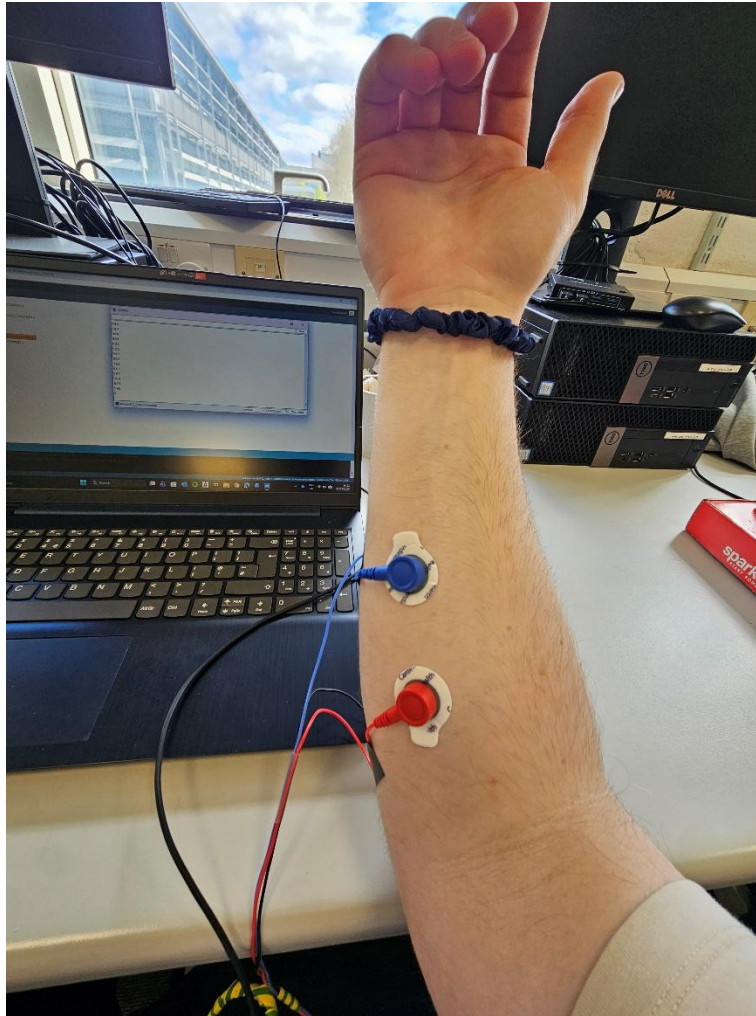


Figure 5.1. EMG acquisition setup showing forearm electrode placement, cable connection, and host laptop used for USB serial monitoring and capture during testing.

5.4 Live Signal Verification

Before structured data capture, the system was tested using a simple Arduino Serial Monitor sketch. This sketch printed the raw analog value from A0 at regular intervals. The purpose of this stage was to confirm whether the sensor, wiring, and electrode placement produced visible changes between rest and contraction.

During early successful live checks, the signal showed clear separation under favourable placement. In one Nano ESP32 bench test, relaxed readings were observed around 250-300 ADC counts and strong contraction readings rose up to approximately 3500 ADC counts. This demonstrated that the hardware chain was capable of producing a large measurable response when electrode contact and placement were favourable.

However, later testing also showed that the signal could become unstable or weak when electrode contact changed. In some trials, resting values drifted or contraction produced only a small change. This variability highlighted the importance of electrode placement, skin preparation, cable strain relief, and repeatable mounting.

These live tests were used as setup verification. Formal numerical claims in this chapter are based on logged CSV sessions rather than live observations alone.

5.5 Guided Rest/Flex Protocol

The main acquisition protocol was the RepeatedFlex5 protocol. Each capture consisted of:

- 7 s settling period
- five repeated cycles, each consisting of 3 s rest and 3 s flex

The participant followed on-screen rest and flex cues generated by the capture script. The script sent marker commands to the Nano ESP32 at each transition so that rest and flex windows could be identified during analysis.

The protocol sequence was:

settle -> rest 1 -> flex 1 -> rest 2 -> flex 2 -> rest 3 -> flex 3 -> rest 4 -> flex 4 -> rest 5 -> flex 5

This ensured that each recorded CSV file contained known rest and contraction windows for later comparison with the processed intent output.

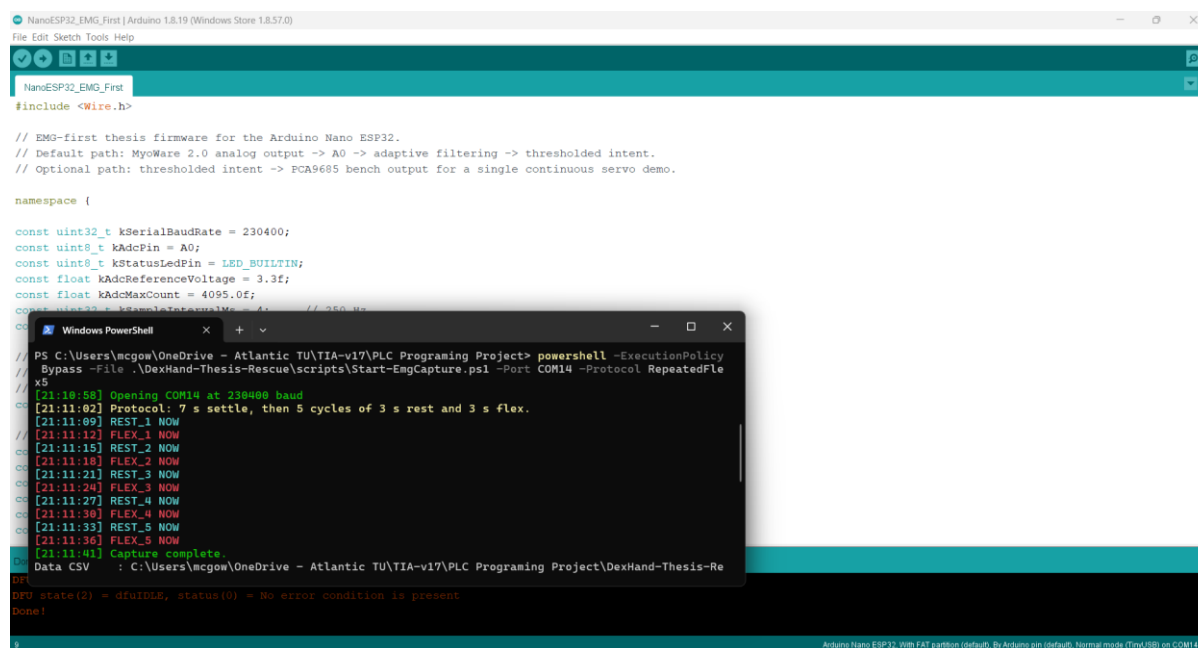


Figure 5.2. Guided RepeatedFlex5 capture workflow showing the Arduino firmware environment and PowerShell protocol script used to issue timed rest/flex cues and log the EMG serial stream.

5.6 Captured Session Data

A structured capture session was successfully recorded using the Nano ESP32 and PowerShell logging workflow. The session produced:

- data CSV file
- marker CSV file
- serial log file
- analysis report
- signal plots

- summary metrics

The strongest saved formal room capture is the final lightly calibrated session:

Table 5.1. Summary metrics for final analysed session *emg-session-20260426-194500*.

The analysis summary for this session was:

Metric	Value
Total valid samples	10401
Duration	41.6 s
Estimated sample rate	250 Hz
ADC range	505 to 583
Peak filtered value	0.02237 V
Mean threshold	0.01957 V
Detected flex onset response	268 ms
Missed flex trials	4
False triggers outside flex windows	0

This session was recorded after a minor threshold adjustment in the firmware. It confirmed that the serial logging and analysis pipeline functioned correctly and that a modest but repeatable contraction response could be recorded under bench testing conditions. One of the five instructed flex trials produced a threshold crossing, giving a detected onset response of 268 ms, with zero false triggers outside the flex windows.

The progression across the main captured sessions is also useful because it shows how the system improved through setup refinement and cautious threshold calibration rather than through false claiming.

Table 5.2 Progression across captured EMG sessions.

Session	Conditions	Peak filtered	Mean threshold	Missed flex trials	False triggers	Interpretation
20260417-211828	Flat lab capture	0.00869 V	0.03000 V	5	0	Logging workflow worked, but no useful rest/flex separation

Session	Conditions	Peak filtered	Mean threshold	Missed flex trials	False triggers	Interpretation
20260426-183627	First structured room capture	0.02085 V	0.03062 V	5	0	Measurable raw flex response, no threshold crossings
20260426-193342	First calibrated room capture	0.02028 V	0.02034 V	5	0	Near-threshold response, missed by only 0.00006 V
20260426-194500	Final light-threshold tweak	0.02237 V	0.01957 V	4	0	Limited binary detection evidence with one true flex detection

5.7 Raw Signal and Feature Response

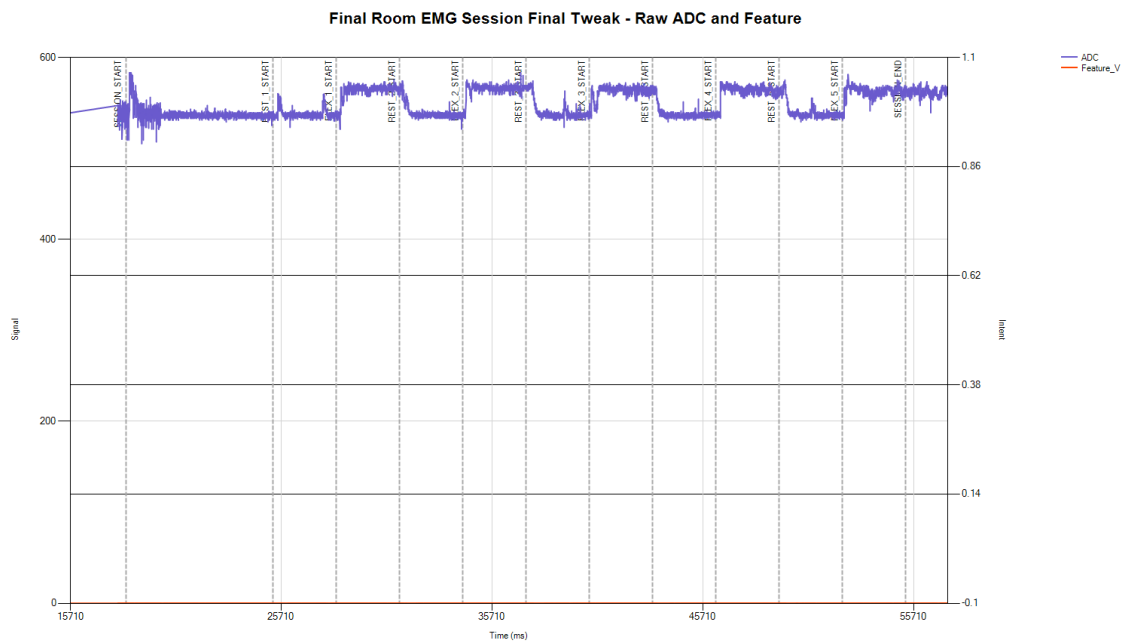


Figure 5.3. Raw ADC response and extracted activation feature for session emg-session-20260426-194500, showing a modest but consistent increase during instructed flex windows.

In the analysed 20260426-194500 final lightly calibrated session, the cleaned raw ADC values ranged from 505 to 583. While the overall signal change was still modest, the signal was more informative than in the earlier flat lab capture. The rest windows remained stable

around the high-530 to low-540 ADC range, while the flex windows rose consistently into the low-to-mid 560 range.

Table 5.3 Window-by-window rest and flex ADC averages for session *emg-session-20260426-194500*.

Trial	Rest average	Flex average	Flex-rest delta
1	537.69	562.67	24.97
2	539.66	564.37	24.71
3	541.19	562.19	21.01
4	538.75	562.47	23.72
5	540.69	562.11	21.41

The mean flex-rest average difference across the five trials was approximately 23.16 ADC counts, with the largest flex peak rising approximately 43.34 ADC counts above the corresponding resting average. This is still not a large control margin, but it is sufficient to support the claim that a measurable raw acquisition response was present during deliberate contraction.

This result was important because it showed that the system could record a consistent directional change in the signal without yet achieving a large enough margin for reliable threshold-based classification.

5.8 Filtered Signal and Threshold Response

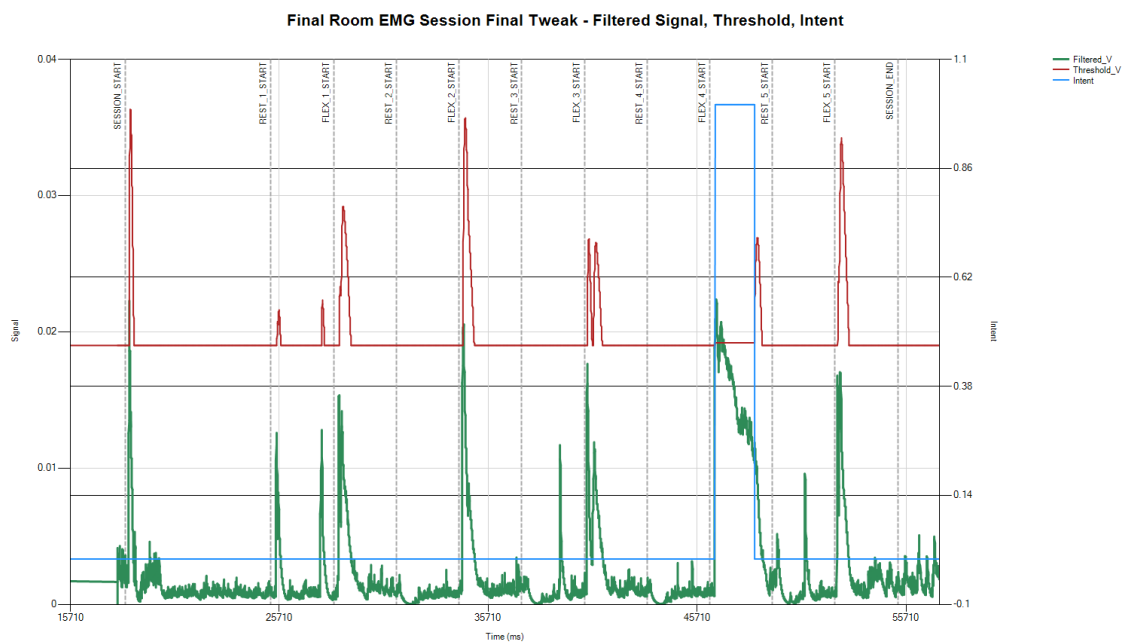


Figure 5.4. Filtered activation signal, adaptive threshold, and binary intent output for session *emg-session-20260426-194500*. One flex window crossed the threshold, while no false triggers occurred outside the instructed flex windows.

For the analysed final lightly calibrated room session, the peak filtered value was 0.02237 V, while the mean threshold was 0.01957 V. This session therefore achieved a limited threshold crossing response, with one flex window producing a binary intent change and a reported onset response of 268 ms. However, four of the five flex windows were still missed, so the result should be interpreted as partial rather than robust detection.

This behaviour suggests that cautious calibration can materially improve detection outcome, but it also shows that the signal available at the electrode interface was still only modestly above the resting band. Under stronger acquisition conditions, the same filtering structure would be more likely to produce more reliable threshold crossings across all five trials. The present result therefore supports limited binary intent-state change while still showing that robust EMG control requires both suitable software thresholds and a stronger, more repeatable biological signal.

5.9 Missed Trials and False Triggers

The final lightly calibrated repeated-flex session produced:

- missed flex trials: 4
- false triggers outside flex windows: 0
- threshold detections inside flex windows: 1

The absence of false triggers indicates that the system did not incorrectly detect contraction during rest. The single detected flex trial shows that the thresholded control chain can register a binary intent-state change under controlled conditions. However, the four missed flex trials show that detection remained inconsistent and that the signal margin was still too small for dependable use.

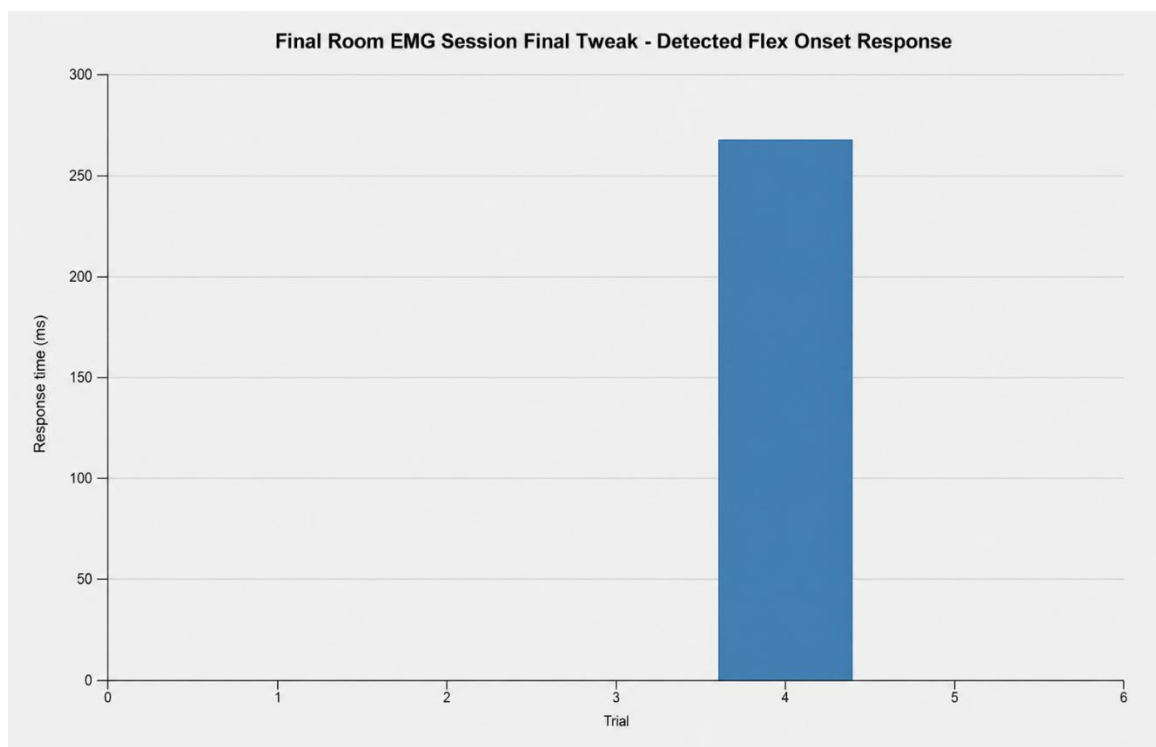


Figure 5.5. Detected flex onset response for session *emg-session-20260426-194500*. Only one instructed flex trial produced a threshold crossing, giving an onset response of approximately 268 ms.

This result is useful because it identifies the main system limitation: reliable classification was not limited by the serial logging or analysis workflow, but by the size and repeatability of the acquired EMG signal relative to the detection threshold. Possible causes include electrode contact quality, placement over a less active muscle region, cable movement, skin impedance, or insufficient signal amplitude under that specific setup.

5.10 Practical Signal Variability Observed During Testing

A major result of the experimental work was the observed variability of low-cost surface EMG acquisition. The system could produce strong live readings under favourable conditions, but it could also become unstable or weak after electrodes were moved, pads were reused, or cable strain changed.

The main observed sources of variability were:

- electrode placement sensitivity
- inconsistent electrode adhesion
- skin preparation differences
- cable movement and strain
- pressure artefacts when pressing electrodes by hand
- changes in forearm posture
- environmental or electrical noise during testing

This variability does not invalidate the project. Instead, it demonstrates an important practical engineering challenge in low-cost EMG systems. Reliable prosthetic control requires not only signal processing algorithms but also repeatable mechanical and skin-contact design.

5.11 Result Claim Summary

The evidence supports the following claims:

- A low-cost EMG acquisition chain was implemented using the MyoWare 2.0 and Arduino Nano ESP32.
- The system successfully streamed and logged timestamped EMG-channel data over USB serial.
- The host-side workflow successfully produced CSV data, protocol markers, plots, and summary statistics.
- Live testing showed that the hardware could produce visible ADC increases during contraction under favourable electrode placement.
- The final analysed structured capture demonstrated limited binary intent detection, with one threshold crossing inside a flex window and zero false triggers outside the flex windows.

The evidence does not support the following claims:

- clinically reliable prosthetic control
- robust multi-gesture classification
- accurate individual finger actuation
- validated IMU or robotic arm control
- repeatable binary intent detection across all sessions

This claim boundary is important because it keeps the thesis aligned with the measured evidence.

5.12 Results Summary

The experimental results show that:

- the MyoWare 2.0 and Nano ESP32 hardware chain can acquire analog EMG-related data
- live testing demonstrated that stronger rest-to-contraction separation is possible under favourable electrode placement
- the structured capture workflow successfully recorded over ten thousand samples in a guided protocol
- the analysis pipeline generated plots, summary metrics, and detection statistics
- the final calibrated room capture showed a modest but consistent raw increase during flex windows
- the final light-threshold tweak produced one threshold crossing inside a flex window with zero false triggers
- binary detection was therefore demonstrated only in a limited and inconsistent form
- electrode placement and contact quality were the dominant practical limitations

The most defensible conclusion from the results is that the project successfully developed the EMG acquisition and analysis workflow, demonstrated a measurable raw EMG response under controlled bench-test conditions, and achieved limited binary intent detection after light calibration. The final testing sequence also showed that the practical bottleneck was not the logging pipeline, but the difficulty of maintaining enough signal margin above rest to produce repeatable threshold crossings. Further work is required to increase the signal margin above rest and to make the live signal consistently robust enough for dependable robotic hand actuation.

Chapter 6 — Discussion

6.1 Overview

This chapter discusses the significance of the EMG acquisition and intent-detection results. The project began with the wider aim of exploring muscle-signal control for a dexterous robotic prosthetic hand. During development, the validated scope was narrowed to the most important first stage of that control chain: acquiring a usable EMG signal and processing it into a control intent.

This narrowing was an appropriate engineering decision. A robotic hand cannot be controlled reliably if the input signal is unstable or poorly understood. By focusing on signal acquisition, filtering, thresholding, and repeatable testing, the project produced a clearer and more defensible contribution than an under-validated full hand-control demonstration.

6.2 Feasibility of Low-Cost EMG Acquisition

The project showed that a low-cost MyoWare 2.0 sensor and Arduino Nano ESP32 can be used to build a practical EMG acquisition workflow. The system was able to stream analog readings, log structured CSV sessions, and generate analysis plots from recorded data. Live bench testing also showed that strong separation between rest and contraction is possible when electrode placement and contact are favourable.

This supports the central project argument that low-cost EMG hardware can provide a useful foundation for assistive or prosthetic control research. The hardware used was accessible, compact, and suitable for student-level prototyping. The processing method was also lightweight enough to run on an embedded microcontroller.

However, the results also showed that successful EMG acquisition is not guaranteed simply by connecting the sensor. Surface EMG depends heavily on the quality of the skin-electrode interface. This makes the mechanical and practical setup just as important as the electronics and software.

6.3 Electrode Sensitivity and Contact Quality

The most significant practical limitation observed during testing was electrode sensitivity. Small changes in electrode position, pressure, cable strain, or skin preparation could produce large changes in signal behaviour. In some cases, pressing on the electrodes caused the signal to drop or jump sharply. This indicated that contact artefacts could dominate the reading if the pads were not seated consistently.

The project therefore demonstrated an important real-world issue in myoelectric control. EMG systems do not operate in ideal laboratory conditions when used as wearable devices. They must cope with moving skin, changing contact pressure, sweat, body hair, cable movement, and user posture. For a prosthetic device, these issues would become even more important because the sensor would need to remain reliable throughout daily activity.

This finding supports the need for improved electrode mounting in future work. A more robust system should use a repeatable strap, sleeve, or socket-integrated electrode mount

rather than loose pads and free-hanging cables. Cable strain relief should also be built into the mechanical design.

This contact sensitivity is one of the most important engineering findings of the project. The project did not simply show that the sensor can be connected to a microcontroller; it showed that the human-sensor interface is a limiting design feature. In future prosthetic use, the electronics would need to be paired with a repeatable physical interface that maintains electrode pressure, reduces cable movement, and places the electrodes over the same muscle region each time.

6.4 Interpretation of Missed Detection Results

The analysed final repeated-flex session successfully captured 10401 valid samples at approximately 250 Hz, showing that the logging workflow was functional. However, the same session detected only one of the five instructed flex trials, while four flex windows were missed. This result should not be interpreted as a complete system failure. Instead, it identifies the weakest part of the current prototype: repeatable acquisition of a strong biological signal.

The result separates the system into two different conclusions:

- the electronic logging and analysis workflow operated correctly
- the electrode/contact condition produced a measurable raw response, but not enough signal margin for reliable threshold detection

This distinction is important. The project succeeded in building the acquisition and analysis infrastructure, but further work is required to make the input signal robust enough for repeatable live control. The missed trials therefore provide useful evidence for the limitations section rather than simply being discarded.

The result also demonstrates why live visual response and logged detection must be treated separately. A live Serial Monitor response can show that the system is capable of reacting to muscle activity, but a formal detection claim requires the logged and analysed data to show threshold crossings inside the flex windows. This distinction prevents the project from overstating its results while still recognising the practical progress made during development.

6.5 Filtering and Thresholding Approach

The filtering and thresholding approach was intentionally simple. Baseline tracking, smoothing, adaptive thresholding, and hysteresis were selected because they are practical for embedded systems and easy to interpret. **This was appropriate for an initial low-cost EMG intent-detection prototype.**

A more complex classifier would not necessarily solve the main problem observed during testing. If the electrode signal is unstable or weak, advanced classification may still fail. The project therefore reinforces a key principle of biomedical sensing: reliable signal acquisition must come before complex signal interpretation.

The binary threshold approach remains useful because it provides a clear first control stage. A future prosthetic hand could use the binary intent output to command a simple open/close

state or to trigger a more advanced control mode. Once robust acquisition is achieved, the system could be extended to proportional control or multi-channel gesture classification.

For this reason, the next technical improvement should not be a more complicated classifier. The first improvement should be a more repeatable acquisition setup, followed by a simple calibration routine that records rest and contraction levels before each session. Only after this stage is reliable would multi-gesture classification or proportional control be justified.

6.6 Mechanical Lessons from the DexHand Prototype

The DexHand platform provided a valuable mechanical target for the EMG control concept. It helped connect the signal-processing work to a visible prosthetic application and supported the project demonstration. However, the mechanical implementation also introduced important design lessons.

Fishing line was considered as a low-cost tendon material. It is cheap, flexible, and easy to obtain, which makes it attractive for open-source robotic hands. However, it was difficult to tension consistently and may reduce repeatability. Tendon slack, friction through printed channels, and inconsistent anchoring can all affect finger motion.

This is an important cost-versus-reliability tradeoff. The purpose of low-cost prosthetic research is not simply to build the cheapest possible device. The purpose is to reduce cost while maintaining enough reliability, control, and usability to help the intended user. For future DexHand development, better tendon material, improved routing, and adjustable tensioning should be investigated.

6.7 Servo and Actuator Limitations

The available 360-degree continuous-rotation micro servos were another limitation. Continuous-rotation servos are useful for speed and direction control, but they are not ideal for accurate finger position control. A dexterous robotic or prosthetic hand would require repeatable finger positioning and controlled tendon displacement. For this reason, true positional micro servos would be more appropriate for future work.

Future actuator selection should prioritise:

- positional control
- torque capacity
- low backlash
- reliable centring
- suitable size and weight
- consistent PWM response
- availability and cost

This finding is important for the project's accessibility goal. Low-cost parts are valuable only if they support reliable function. If a cheaper actuator prevents precise control, it may reduce the usefulness of the future prosthetic system.

6.8 IMU and Robotic Arm De-Prioritisation

IMU sensing and robotic arm control were considered during the early project concept. These remain relevant future extensions, especially for tracking arm orientation or supporting teleoperation. However, they were deliberately de-prioritised in the validated project scope.

This decision improved the quality of the thesis by preventing the work from becoming too broad. A complete hybrid EMG-IMU robotic arm system would require additional hardware, sensor fusion, calibration, mapping, and safety validation. Attempting to include these elements without enough test data would weaken the thesis.

By treating IMU and robotic arm control as future work, the project remains honest about what was actually implemented while still showing a clear path for expansion.

6.9 Relevance to Prosthetic and Assistive Technology

The project is relevant to prosthetic and assistive technology because it addresses the first step in myoelectric control: interpreting muscle activity as user intent. Even a simple rest-versus-contraction detector can form the basis of a future open/close prosthetic control strategy.

The low-cost emphasis is also important. Advanced prosthetic systems can be expensive, difficult to maintain, and inaccessible to many users. A low-cost sensing and control pipeline may support educational research, early-stage prototyping, rehabilitation exercises, or future affordable assistive devices.

However, the system should not be described as clinically ready. It was tested as a bench-level prototype with one participant. Further validation would be required across different users, skin conditions, electrode placements, fatigue levels, and daily movement scenarios.

6.10 Overall Discussion Summary

The main contribution of this project is the development and evaluation of a low-cost EMG acquisition and intent-detection workflow. The work showed that the hardware and software chain can be built using accessible components and that structured data capture and analysis can be performed. It also showed that reliable EMG control depends strongly on electrode placement and physical setup.

The most important lesson is that future prosthetic control is not only a software problem. It is a combined sensing, mechanical, electrical, and human-interface problem. The project therefore provides a realistic foundation for future development while identifying the practical improvements required before controlled DexHand actuation can be achieved.

Chapter 7 — Conclusion and Future Work

7.1 Conclusion

This project investigated the development of a low-cost EMG acquisition and intent-detection system for future prosthetic or assistive robotic hand control. The work was motivated by the need for accessible prosthetic technologies that can interpret human muscle activity and translate it into useful robotic movement.

The final validated system used a SparkFun MyoWare 2.0 muscle sensor and an Arduino Nano ESP32 to acquire forearm muscle activity. The embedded firmware sampled the analog signal, estimated a baseline, extracted an activation feature, applied digital smoothing, and compared the filtered output against an adaptive threshold. A host-side logging and analysis workflow was also developed to capture structured rest/flex trials and generate plots and summary metrics.

The project demonstrated that a low-cost EMG acquisition workflow can be implemented using accessible components. It also demonstrated that live muscle-signal response is possible under favourable electrode placement and contact conditions. However, the results showed that surface EMG is highly sensitive to electrode placement, skin preparation, cable movement, and contact quality. These factors limited the repeatability of the signal and affected the success of threshold-based detection during structured capture.

The final analysed session produced one threshold crossing during an instructed flex window, with four missed flex trials and zero false triggers, demonstrating limited but not repeatable binary intent detection.

The DexHand platform was developed as the target robotic hand context for the project. It provided a visible mechanical direction for future prosthetic control work. However, full DexHand actuation was not treated as the primary validated result. This was an important scope decision because the most defensible contribution of the project was the EMG acquisition and processing pipeline.

Overall, the project achieved its revised core aim of developing and evaluating a low-cost EMG signal pathway suitable as a foundation for future prosthetic or assistive robotic control. The work also identified the practical improvements required before the system could become a reliable real-time hand controller.

7.2 Key Outcomes

The main outcomes of the project were:

- development of a MyoWare 2.0 and Nano ESP32 EMG acquisition system
- implementation of real-time baseline tracking, filtering, thresholding, and binary intent detection
- creation of a guided rest/flex capture protocol
- successful serial logging of EMG sessions to CSV files
- generation of analysis plots and summary metrics
- identification of electrode placement and contact quality as major limitations

- development of a DexHand mechanical prototype as the target application platform
- identification of fishing line and continuous-rotation servo limitations for controlled finger movement

These outcomes support the conclusion that low-cost EMG acquisition is feasible at prototype level, but reliable prosthetic control requires careful attention to both signal quality and mechanical implementation.

7.3 Project Limitations

Several limitations should be recognised.

First, the project was tested as a single-participant bench study. The results should not be interpreted as clinical evidence or as proof of prosthetic readiness for amputee users.

Second, the validated system used a single EMG channel and binary intent detection. It did not validate multi-channel gesture recognition, proportional grip control, or independent finger movement.

Third, the EMG signal was sensitive to electrode placement and contact quality. This affected repeatability and made live demonstration challenging.

Fourth, the DexHand actuation stage was not fully validated. The available continuous-rotation servos were not ideal for accurate finger position control, and tendon routing using fishing line introduced practical issues with tensioning and repeatability.

Finally, IMU sensing and robotic arm teleoperation were not implemented as validated parts of the final system. They remain future extensions rather than completed outcomes.

7.4 Future Work

7.4.1 Improved Electrode Mounting

Future work should prioritise a more repeatable electrode mounting method. A strap, sleeve, or printed forearm mount could reduce movement artefacts and improve contact consistency. Cable strain relief should also be included so that the electrode pads are not pulled during movement.

7.4.2 Multi-Channel EMG

The system should be expanded from one EMG channel to multiple channels. Multi-channel acquisition would allow different forearm muscles to be compared and could support more advanced gesture recognition. This would also reduce reliance on a single electrode pair.

7.4.3 Improved Calibration and Thresholding

Future firmware should include a guided calibration stage that records rest and contraction levels before setting thresholds. This would allow thresholds to be tailored to the user and session. Adaptive algorithms could also account for fatigue and gradual baseline drift.

7.4.4 DexHand Actuation

Future work should integrate the EMG intent output with the DexHand actuation system. The PCA9685 servo driver could be used to control multiple servos over I2C, but the servos should be powered from a separate supply with a shared ground reference. Safety limits should be included to prevent excessive tendon tension or finger travel.

7.4.5 Improved Tendon and Servo Selection

Fishing line should be replaced or supplemented if more repeatable mechanical control is required. Alternative tendon materials, better anchoring methods, and adjustable tensioning mechanisms should be investigated. Future versions should also use positional micro servos rather than continuous-rotation servos for controlled finger movement.

7.4.6 Optional IMU Fusion

IMU sensing could be added in a later version to measure wrist or forearm orientation. This would allow EMG to detect grasp intent while IMU data provide movement context. However, IMU integration should only be added after the EMG signal path is stable and repeatable.

7.5 Final Statement

The project shows that affordable EMG hardware can provide a foundation for assistive robotic control, but it also shows that real-world reliability depends on careful integration of sensors, electronics, software, and mechanics. The next stage of development should focus on making the signal interface more repeatable and then using that improved intent signal to drive controlled DexHand movement.

The long-term goal is not simply to produce a low-cost robotic hand. The goal is to develop a low-cost system that remains reliable, controllable, and useful enough to support people who have lost limb function or require assistive technology.

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Appendices

Appendix A - Supporting Code and Commands

This appendix contains supporting implementation details for the EMG acquisition workflow. Long code listings and commands are placed here rather than in the main thesis body so that the main chapters remain focused on method, results, and discussion.

A.1 Basic Live Signal Test

The following Arduino sketch was used during early live testing to confirm whether the Nano ESP32 was receiving a changing analog signal from the MyoWare sensor.

```
void setup() {  
    Serial.begin(115200);  
  
    delay(1000);  
  
    analogReadResolution(12);  
}  
  
void loop() {  
    int emg = analogRead(A0);  
  
    Serial.println(emg);  
  
    delay(20);  
}
```

This sketch does not filter the signal. It simply prints the raw analog reading from A0, which made it useful for checking electrode placement, wiring, and basic sensor response.

Appendix B - EMG Logging Firmware Summary

The final Nano ESP32 firmware sampled the MyoWare ENV output on analog input A0. The firmware used:

- baud rate: 230400
- sample interval: approximately 4 ms
- estimated sample rate: approximately 250 Hz
- ADC resolution: 12-bit
- input mode: MyoWare envelope output

Each CSV row contained:

sample_ms,adc,volts,baseline_v,feature_v,filtered_v,threshold_v,noise_v,intent,state_change,marker

The firmware implemented:

- baseline tracking
- positive activation feature extraction
- fast and slow smoothing
- adaptive thresholding
- hysteresis-based binary intent detection
- protocol marker support

The full firmware file is stored in the project workspace at:

DexHand-Thesis/firmware/NanoESP32_EMG_First/NanoESP32_EMG_First.ino

The file path is included to identify the implemented firmware version used during testing; the main thesis chapters summarise the operation rather than reproducing the full source code.

Appendix C - Guided Capture Command

The structured rest/flex protocol was captured using the PowerShell logging script.

Example command:

```
cd "C:\Users\mcgow\OneDrive - Atlantic TU\TIA-v17\PLC Programing Project"
```

```
powershell -ExecutionPolicy Bypass -File .\DexHand-Thesis\scripts\Start-EmgCapture.ps1 -Port COM14 -Protocol RepeatedFlex5
```

The COM port was checked using:

```
[System.IO.Ports.SerialPort]::GetPortNames()
```

The capture script generated:

- data CSV
- markers CSV
- serial log

Appendix D - Rest/Flex Protocol

7 s settle

Then five repeated cycles of:

3 s rest

3 s flex

This provided five repeated contraction windows for analysis.

Appendix E - Analysis Command

The newest captured session was analysed using:

```
$p = Get-ChildItem '.\DexHand-Thesis\data\raw\emg-session-*.csv' | Where-Object {
$_.Name -notlike '*.markers.csv' } | Sort-Object LastWriteTime -Descending |
Select-Object -First 1
```

```
$m = Get-ChildItem '.\DexHand-Thesis\data\raw\emg-session-*.markers.csv' | Sort-
Object LastWriteTime -Descending | Select-Object -First 1
```

```
powershell -ExecutionPolicy Bypass -File .\DexHand-Thesis\scripts\Analyze-
EmgSession.ps1 -InputCsv $p.FullName -MarkersCsv $m.FullName -Title 'Real EMG
Session'
```

The analysis script generated:

- raw/feature plot
- filtered/threshold/intent plot
- latency bar chart
- latency table
- summary CSV
- Markdown analysis report

Appendix F - Hardware Wiring Summary

Main EMG acquisition wiring:

MyoWare VIN -> Nano ESP32 3V3

MyoWare GND -> Nano ESP32 GND

MyoWare ENV -> Nano ESP32 A0

Future servo driver wiring concept:

Nano ESP32 SDA/SCL -> PCA9685 I2C

External 5 to 6 V supply -> PCA9685 V+

Nano ESP32 GND -> PCA9685 GND

Servo outputs -> DexHand servos

The servo driver concept requires a separate servo power supply and common ground. The Nano ESP32 should not be used to power multiple servos directly.

Appendix G - Troubleshooting Notes

The following practical issues were observed during development:

- loose jumper wires caused unstable readings
- soldered headers improved connection reliability
- reused electrode pads reduced signal quality
- pressing electrodes by hand caused contact artefacts
- cable strain affected electrode contact
- the Arduino Serial Monitor had to be closed before PowerShell capture
- Nano ESP32 upload issues could require bootloader mode using a double reset
- continuous-rotation servos were unsuitable for precise finger positioning

These observations informed the limitations and future work sections of the thesis.

Appendix H - Design and Source Attributions

The DexHand mechanical platform used in this project was based on externally available CAD/STL design files and supporting build resources. These files were used as the basis for 3D printing and assembling the experimental hand prototype. The DexHand platform was included as a low-cost mechanical target for future EMG-driven actuation and was not presented as an original medical device design.

Any DexHand CAD models, STL files, build guides, or YouTube tutorials used during the project were treated as external design/support resources and are credited to their original creators. These resources supported mechanical understanding, printing, assembly, or demonstration planning, while the validated thesis contribution remained the EMG acquisition, logging, filtering, and intent-detection workflow.

External resources used:

- DexHand CAD/STL source: Rob Knight / The Robot Studio, *V1.0-Dexhand GitHub Repository*, available at: <https://github.com/TheRobotStudio/V1.0-Dexhand>
- DexHand CAD model reference: Rob Knight / The Robot Studio, *DexHand Onshape CAD model*, linked from the V1.0-Dexhand GitHub repository, available at: <https://cad.onshape.com/documents/16c809d35214bd31c5b0324f/w/56b2fb82f7bdce505771766f/e/df78e66574dc7dbd45e4826c>
- DexHand project information and assembly resources: *DexHand.org*, available at: <https://www.dexhand.org/>
- DexHand build/tutorial videos: Rob Knight / The Robot Studio, *DexHand V1 Assembly Playlist*, available at: https://youtube.com/playlist?list=PLy7gxZH9jzfQB2YJvAl5Y0FLJvxzb_83F
- SparkFun MyoWare 2.0 hardware documentation: SparkFun Electronics, *Getting Started with the MyoWare 2.0 Muscle Sensor Ecosystem*, available at: _____

<https://learn.sparkfun.com/tutorials/getting-started-with-the-myoware-20-muscle-sensor-ecosystem>

- Arduino Nano ESP32 hardware documentation: Arduino, *Nano ESP32 Documentation*, available at: <https://docs.arduino.cc/hardware/nano-esp32>
- PCA9685 servo driver documentation: Adafruit, *16-Channel PWM/Servo Driver - PCA9685 Guide*, available at: <https://learn.adafruit.com/16-channel-pwm-servo-driver>

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